# Circuit Theorems Assignment

ECE2107 Circuits & Systems · assignment marathon · built so far: Part A Superposition (5.1–5.14), Thevenin & Norton (5.15–5.34), maximum power (5.35–5.44)

## How to use this

Every question follows the same order:

1. **Question**, with the figure taken straight from the book.
2. **Try it first** (yellow). Cover the rest of the question and solve it.
3. **Hint: what's new** (orange). Only when the question brings an idea you haven't met earlier in this PDF. If you're stuck, read it and try again.
4. **Solution** in compact assignment form: Given → To find → each step with its intermediate values → boxed answer. Copy it into your submission, or compare it line by line with your attempt to find the step that went wrong.
5. **Side note** (grey) or **book error** (red), only when there is something worth knowing.

The tag next to each question number shows how much teaching it carries: **teach** full explanation · **new idea** one new thing · **practice** plain solution.

If you couldn't solve one, just tell me its number. Nothing else needs marking.

## What's in it (full version)

| Part | Book | Questions | Count |
|---|---|---|---|
| A | Ghosh & A K Chakraborty, *Circuit Theory and Networks*, ch. 5 (**solved**) | Superposition 5.1–5.14 · Thevenin/Norton 5.15–5.34 · Max power 5.35–5.44 | 44 |
| B | Alexander & Sadiku, *Fundamentals of Electric Circuits* 5e (**unsolved**) | Practice problems + end-of-chapter: ch. 4 (superposition, source transformation, Thevenin/Norton, max power), ch. 10 (the same in AC), ch. 11.3 (AC max power) | 107 + practice problems |

Scope follows the official MUJ ECE2107 syllabus: superposition, Thevenin's and Norton's theorems, maximum power transfer, DC and AC, plus source transformation.

**Checking.** Every answer was worked out again and confirmed by a full circuit solve with all sources on (`src/verify.py`), including each case's partial answer. Matches with the book are marked "= book". Where the book is wrong, the correct answer is given and a red box explains the book's slip.

# Part A · Chakraborty (solved problems)

Ghosh & A K Chakraborty, *Circuit Theory and Networks* 4e, ch. 5 "Network Theorems", Solved Problems

## A · Superposition theorem

KVL, KCL, Ohm's law (you have these) → Nodal / mesh: all sources at once → **Superposition: one source at a time** → Thevenin / Norton (next)

**Where this sits.** You can already solve any circuit with nodal or mesh analysis, taking every source at once. Superposition breaks the same job into smaller circuits, **one source at a time**, which is often quicker by hand. It is also the reason Thevenin's theorem works, which comes next.

**Three terms first.**

- **Linear circuit:** made of R, L, C, linear dependent sources and independent sources. Doubling every source doubles every current and voltage.
- **Independent source:** a fixed voltage or current (a battery, a 2 A source, 10 cos 4t).
- **Dependent source** (diamond symbol): its value follows some other voltage or current in the circuit, for example $2V_x$.

**Statement.** In a linear circuit with several independent sources, any current or voltage is the **algebraic sum** of the values caused by **each independent source acting alone**.

**Switching a source off** means setting its value to zero:

| Source | Switched off it becomes | Why |
|---|---|---|
| Ideal voltage source | **short circuit** | 0 V across it |
| Ideal current source | **open circuit** | 0 A through it |
| Dependent source | **never switched off** | it is not a source on its own; it follows the circuit |

Superposition adds currents and voltages only, **not power** (P goes as $I^2$, and $(I' + I'')^2 \neq I'^2 + I''^2$).

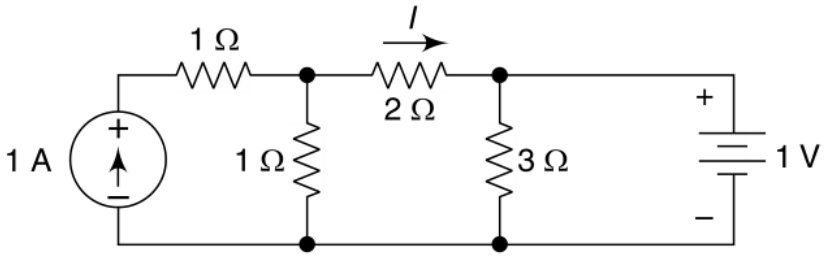
---

### A1 · Problem 5.1

Superposition · DC **teach**

**QUESTION**

Find the current I in the circuit shown, using the superposition theorem.



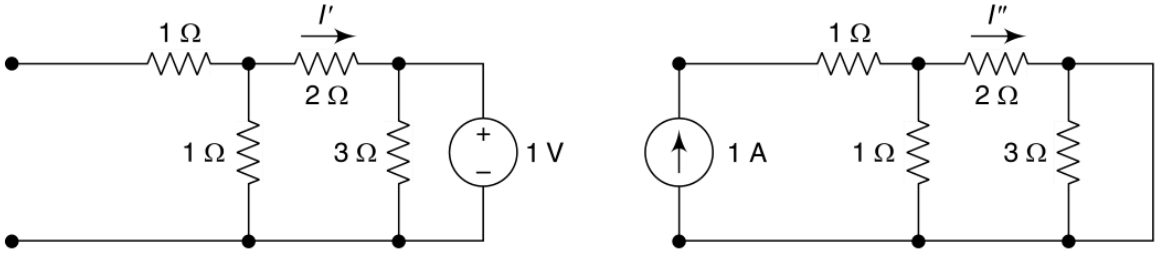

I flows left to right through the 2 Ω resistor.

**TRY IT FIRST**

Cover everything below. Draw two circuits, 1 V alone and 1 A alone, find the current in the 2 Ω in each, and decide each one's sign before adding.

**HINT: WHAT'S NEW**

- **The whole method:** two sources, so solve **two small circuits** and add.
- **Switching off:** the 1 V source becomes a short; the 1 A source becomes an open.
- **Signs:** measure each part against the arrow drawn for I. A part flowing the other way is negative.




(i) 1 V acting alone (1 A opened) · (ii) 1 A acting alone (1 V shorted)

**Given:** 1 A source, 1 V source; 1 Ω (series), 1 Ω (shunt), 2 Ω, 3 Ω.

**To find:** I through the 2 Ω (left → right).

**Case (i): 1 V alone** (1 A opened). The series 1 Ω carries nothing. The 3 Ω sits directly across the 1 V, so it changes nothing else. The 1 V drives current through 2 Ω + 1 Ω (shunt):

$$\text{current} = \frac{1}{2+1} = \frac{1}{3}\ \text{A (right → left)} \qquad \Rightarrow \qquad I' = -\frac{1}{3}\ \text{A}$$

**Case (ii): 1 A alone** (1 V shorted, so the 3 Ω is shorted out). The 1 A divides between the 1 Ω shunt and the 2 Ω:

$$I'' = 1 \times \frac{1}{1+2} = \frac{1}{3}\ \text{A (left → right)}$$

**Superposition:** $I = I' + I'' = -\frac{1}{3} + \frac{1}{3} = 0$

**ANSWER**

**I = 0 A** (= book)

**WHY ZERO, AND TWO SHORTCUTS**

- Node check: $1 = \frac{V_A}{1} + \frac{V_A - 1}{2}$ gives $V_A = 1$ V, the same as the right-hand node, so no current flows in the 2 Ω.
- An element **in series with an ideal current source** never changes any other current (the left 1 Ω here).
- An element **in parallel with an ideal voltage source** never changes any other current (the 3 Ω here).

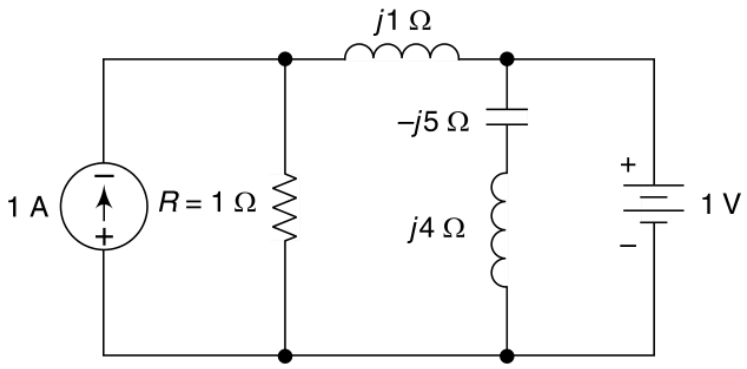
---

## A2 · Problem 5.2

Superposition · AC **teach**

**QUESTION**

Calculate the voltage V across the resistor R by using the superposition theorem.



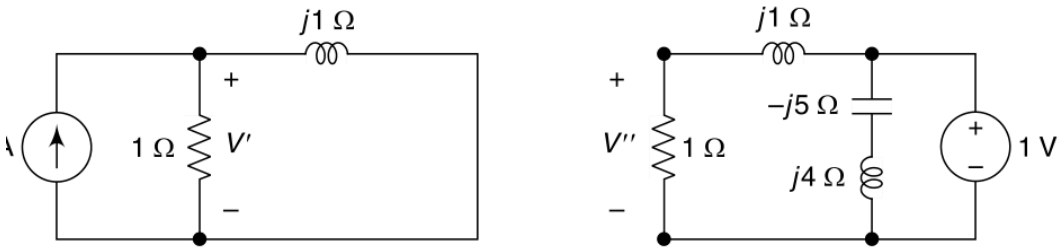

The book draws +/− signs inside the current source. Read it by its arrow: 1 A pushed up into the top node.

**TRY IT FIRST**

Which elements drop out in each case? Aim for V′ and V″ in the form $\frac{\text{something}}{1 + j}$.

**HINT: WHAT'S NEW**

- **AC (phasor) circuit.** j1 Ω is an inductor, −j5 Ω a capacitor. Superposition is unchanged; the arithmetic uses complex numbers.
- **Complex dividers:** the same formulas as for resistors, with Z in place of R.
- The −j5 + j4 branch is **across the 1 V source** (5.1's shortcut), so it never matters.




(i) 1 A acting alone · (ii) 1 V acting alone

**Given:** 1 A, 1 V; R = 1 Ω, j1 Ω, −j5 Ω, j4 Ω.

**To find:** V across R (+ at top).

**Case (i): 1 A alone** (1 V shorted; this also shorts out the −j5 + j4 branch). R ∥ j1 across the 1 A:

$$V' = 1 \times \frac{1 \times j1}{1 + j1} = \frac{j}{1 + j}\ \text{V}$$

**Case (ii): 1 V alone** (1 A opened). j1 and R in series across 1 V:

$$I'' = \frac{1}{1 + j} \qquad \Rightarrow \qquad V'' = 1 \times I'' = \frac{1}{1 + j}\ \text{V}$$

**Superposition:** $V = \frac{j}{1 + j} + \frac{1}{1 + j} = 1\ \text{V}$

**ANSWER**

**V = 1∠0° V** (= book)

**CHECK**

Node equation at the top node: $1 = \frac{V}{1} + \frac{V - 1}{j1}$ gives V(1 + j) = 1 + j, so V = 1 V ✓.

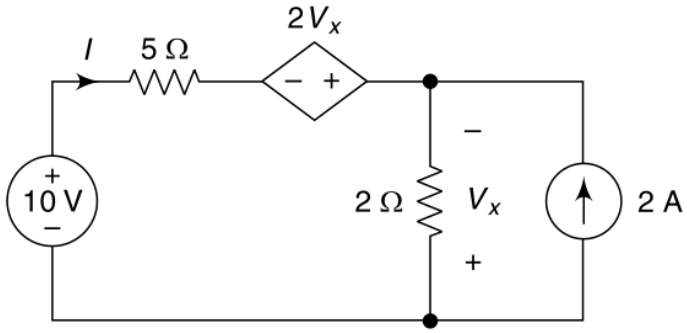
---

## A3 · Problem 5.3

Superposition · dependent source **teach**

**QUESTION**

Use the superposition theorem on the circuit shown to find I.



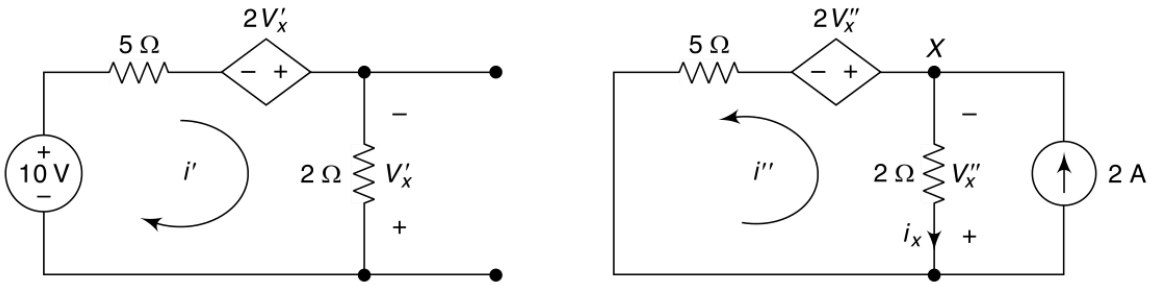

$2V_x$ is a voltage-controlled voltage source. $V_x$ is across the 2 Ω with + at the bottom.

**TRY IT FIRST**

Write $V_x'$ in terms of the loop current in case (i). For case (ii), write one KCL at the top of the 2 Ω and one KVL around the left loop. Expect elevenths.

**HINT: WHAT'S NEW**

- **Dependent source:** it **stays in every case**. Only the 10 V and the 2 A take turns.
- **Its value changes per case:** $2V_x'$ in case (i), $2V_x''$ in case (ii). Re-find $V_x$ in each sub-circuit.
- **Polarity of $V_x$:** + is at the **bottom** of the 2 Ω. Most wrong answers to this question come from this sign.




(i) 10 V alone (2 A opened) · (ii) 2 A alone (10 V shorted); the dependent source stays in both

**Given:** 10 V, 2 A, 5 Ω, 2 Ω, VCVS $2V_x$.

**To find:** I (out of the 10 V source, through the 5 Ω).

**Case (i): 10 V alone.** One loop, current i′ clockwise (= direction of I), flowing down the 2 Ω, so $V_x' = -2i'$.

$$10 + 2V_x' = 5i' + 2i' \quad \Rightarrow \quad 10 - 4i' = 7i' \quad \Rightarrow \quad I' = i' = \frac{10}{11}\ \text{A}$$

**Case (ii): 2 A alone.** At node X: 2 A splits into $i_x$ (down the 2 Ω, so $V_x'' = -2i_x$) and i″ (leftward through the dependent source and 5 Ω).

$$\text{KCL at X:} \qquad 2 = -\frac{V_x''}{2} + i'' \qquad \ldots(1)$$

$$\text{Left loop:} \qquad 5i'' + 3V_x'' = 0 \qquad \Rightarrow \qquad i'' = -\frac{3}{5}V_x'' \qquad \ldots(2)$$

(X sits at $-V_x''$; the dependent source drops another $2V_x''$, so the 5 Ω has $-3V_x''$ across it.)

$$\text{(2) in (1):} \qquad 2 = -\frac{11}{10}V_x'' \qquad \Rightarrow \qquad V_x'' = -\frac{20}{11}\text{ V}, \qquad i'' = \frac{12}{11}\text{ A}$$

$i''$ runs anticlockwise, against I, so **$I'' = -12/11$ A**.

**Superposition:** $I = \frac{10}{11} - \frac{12}{11} = -\frac{2}{11}$ A

**ANSWER**

**$I = -2/11$ A ≈ −0.182 A** (= book). The minus sign means the current actually flows into the + of the 10 V source.

**NEVER SWITCH OFF A DEPENDENT SOURCE**

Shorting $2V_x$ as if it were independent gives a wrong answer. It stays in; only its value changes from case to case.

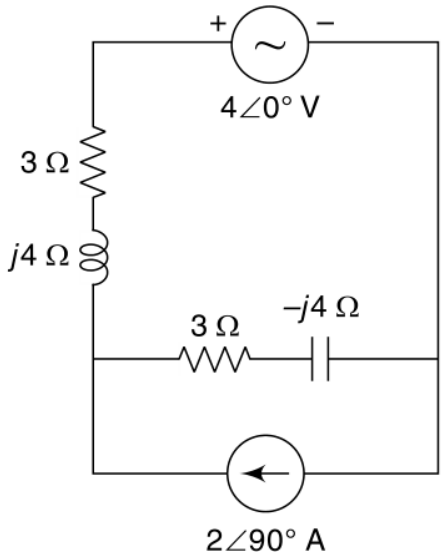
---

## A4 · Problem 5.4

Superposition · AC **new idea**

**QUESTION**

Determine the current in the capacitor branch by the superposition theorem.



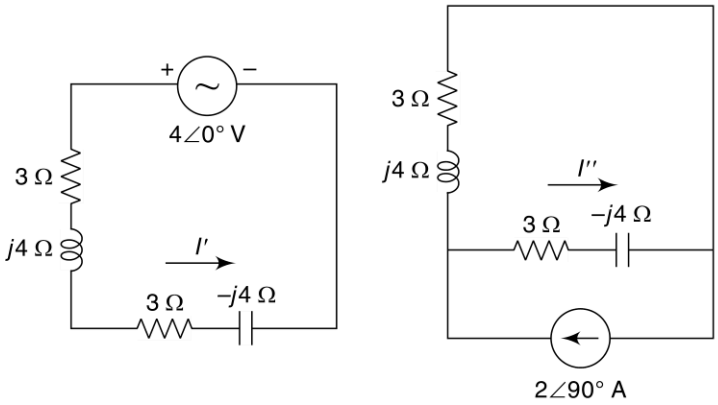

$Z_1 = 3 + j4\ \Omega$ (left branch); $Z_2 = 3 - j4\ \Omega$ (capacitor branch); current source $2\angle 90°$ A.

**TRY IT FIRST**

Which way does each source see $Z_1$ and $Z_2$: in series, or in parallel?

**HINT: WHAT'S NEW**

- **Source with an angle:** $2\angle 90°$ A = j2 A. Convert to rectangular form before multiplying.
- **Polar form with a negative real part:** a calculator's $\tan^{-1}$ lands in the wrong quadrant; add 180°.




(i) $4\angle 0°$ V alone · (ii) $2\angle 90°$ A alone

**Given:** $4\angle 0°$ V, $2\angle 90°$ A, $Z_1 = 3 + j4\ \Omega$, $Z_2 = 3 - j4\ \Omega$.

**To find:** I in the capacitor branch (left → right).

**Case (i): $4\angle 0°$ V alone** (current source opened). $Z_1$, $Z_2$ in series:

$$I' = \frac{4\angle 0°}{(3 + j4) + (3 - j4)} = \frac{2}{3}\angle 0°\ \text{A}$$

**Case (ii): 2∠90° A alone** (voltage source shorted). $Z_1 \parallel Z_2$ across the source:

$$I'' = j2 \times \frac{3 + j4}{6} = -\frac{4}{3} + j1 \text{ A}$$

**Superposition:** $I = \frac{2}{3} - \frac{4}{3} + j1 = -\frac{2}{3} + j1$ A

$$|I| = \sqrt{((2/3)^2 + 1)} = 1.20 \text{ A}, \qquad \theta = 180° - \tan^{-1}(1.5) = 123.7°$$

**ANSWER**

**I = 1.2∠123.7° A** (= book)

**SIDE NOTE**

$Z_1 + Z_2 = 6\ \Omega$ is purely real (+j4 and −j4 cancel), which is why both cases are clean. Equal and opposite reactances in series act like a short: **series resonance**, met later.

---

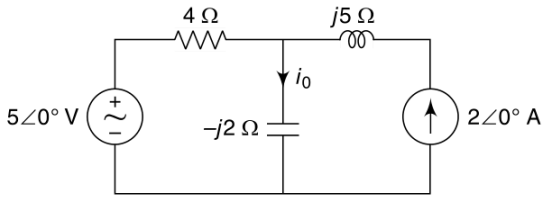
## A5 · Problem 5.5

Superposition · AC, DC + AC **teach**

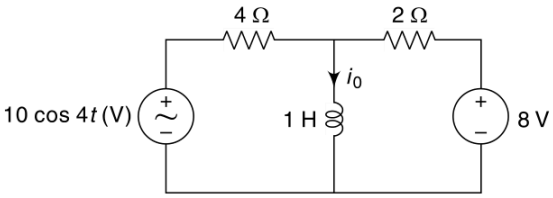
**QUESTION**

Find the current $i_0$ using the superposition theorem, in each of circuits (a), (b) and (c).

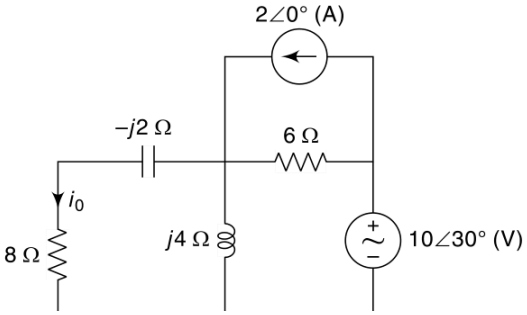
(a)




(b)




(c)



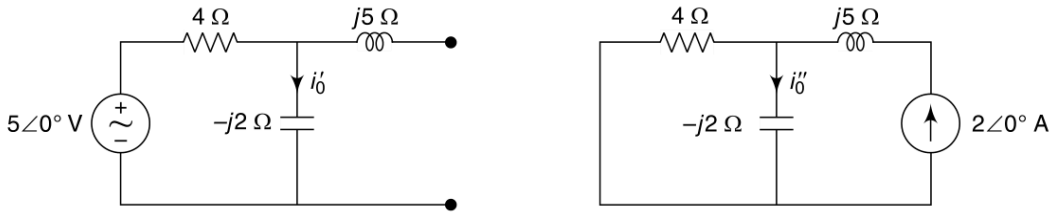
**TRY IT FIRST**

(a) is quick practice. In (b), stop and ask: what does an inductor look like to the 8 V **DC** source? In (c), redraw the circuit after switching off each source before you calculate anything.

**HINT: WHAT'S NEW**

- **(b) Two different frequencies (DC and 10 cos 4t).** Each case is solved **at its own frequency**: at DC ($\omega = 0$), $X_L = \omega L = 0$, so the **inductor is a short**; at $\omega = 4$, it is j4 Ω. The two answers are added **in the time domain**, never as phasors, because phasors only add when they share one frequency.
- **(c) Redraw after switching off.** Once the voltage source is shorted, the 6 Ω and the current source both run to ground and everything becomes parallel.

**(a) Given:** 5∠0° V, 2∠0° A, 4 Ω, −j2 Ω, j5 Ω. **To find:** $i_0$ (down the capacitor).




(i) voltage source alone · (ii) current source alone

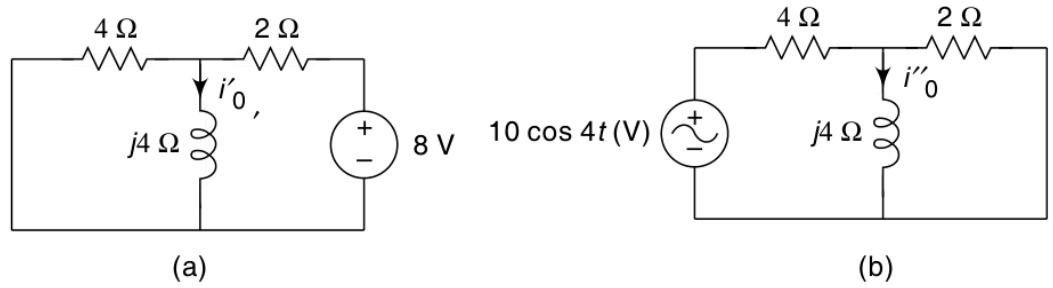
$$5\angle 0° \text{ alone:} \qquad i_0' = \frac{5}{4 - j2} = 1 + j0.5 \text{ A}$$

$$2\angle 0° \text{ alone (j5 is in series with the source, so irrelevant):} \qquad i_0'' = 2 \times \frac{4}{4 - j2} = 1.6 + j0.8 \text{ A}$$

**ANSWER**

**(a) $i_0 = 2.6 + j1.3 = 2.91\angle 26.57°$ A** (= book, 2.9∠26.56° A)

**(b) Given:** 10 cos 4t V, 8 V DC, 4 Ω, 1 H, 2 Ω. **To find:** $i_0(t)$ (down the inductor).




The book's sub-circuits (a) DC alone, (b) AC alone. In (a) the inductor must be read as a short circuit.

**8 V DC alone** (AC source shorted; $\omega = 0$, so the inductor is a short):

$$i_0' = \frac{8}{2} = 4\ \text{A} \qquad \text{(the 4 Ω has 0 V across it)}$$

**10 cos 4t alone** (8 V shorted; ω = 4 rad/s, $Z_L$ = j4 Ω):

$$Z = 4 + \frac{j4 \times 2}{2 + j4} = 5.6 + j0.8\ \Omega, \qquad I = \frac{10\angle 0°}{Z} = 1.75 - j0.25\ \text{A}$$

$$i_0'' = I \times \frac{2}{2 + j4} = 0.25 - j0.75 = 0.791\angle -71.57°\ \text{A}$$

**Superposition (time domain):**

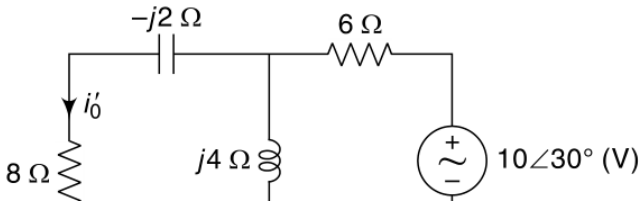
**ANSWER**

**(b) $i_0(t) = 4 + 0.791 \cos(4t - 71.57°)$ A**

**BOOK ERROR IN 5.5(B)**

The book treats the inductor as j4 Ω for the **DC** source too, and then adds the DC and AC results as if both were phasors, printing 2.63∠−63.58° A. Both steps are wrong: at DC an inductor is a short, and quantities at different frequencies cannot be added as phasors. The answer above was confirmed by a node solve of each case.

**(c) Given:** 10∠30° V, 2∠0° A, 8 Ω, −j2 Ω, j4 Ω, 6 Ω. **To find:** $i_0$ (down the 8 Ω).

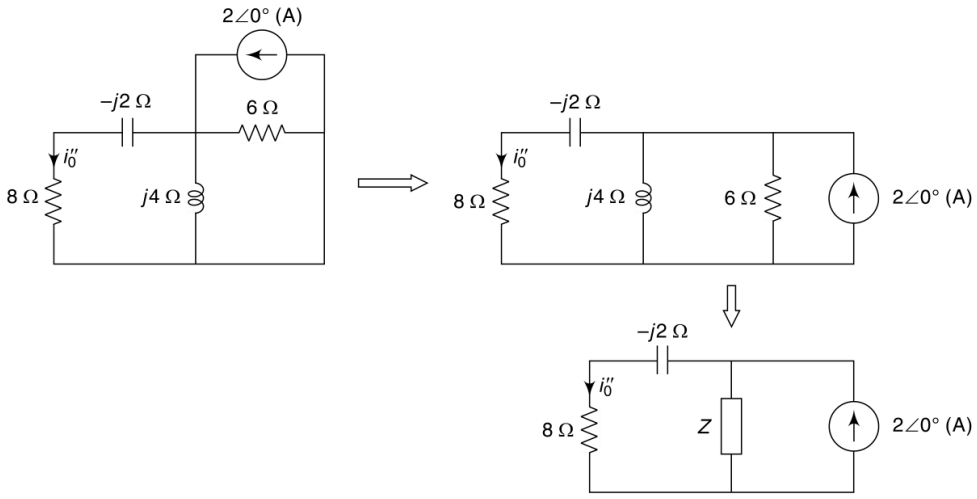



(i) voltage source alone

**10∠30° V alone** (current source opened):

$$Z = 6 + j4 \parallel (8 - j2) = 6 + (1.882 + j3.529) = 7.882 + j3.529\ \Omega, \qquad I = \frac{10\angle 30°}{Z} = 1.158\angle 5.88°\ \text{A}$$

$$i_0' = I \times \frac{j4}{(8 - j2) + j4} = I \times \frac{j4}{8 + j2} = 0.562\angle 81.84°\ \text{A}$$




(ii) current source alone, redrawn: the 6 Ω and the source now both go to ground

**2∠0° A alone** (voltage source shorted). Z = j4 ∥ 6 = 1.846 + j2.769 Ω:

$$i_0'' = 2 \times \frac{Z}{(8 - j2) + Z} = \frac{j24}{28 + j22} = 0.674\angle 51.84^\circ \text{ A}$$

$$i_0 = (0.080 + j0.556) + (0.416 + j0.530) = 0.496 + j1.086 \text{ A}$$

**ANSWER**

**(c) $i_0 = 1.19\angle 65.45^\circ$ A**

**BOOK ERROR IN 5.5(C)**

The book prints 0.81∠39.38° A. Two slips: in case (i) it divides the current with (8 − j2)/(8 + j2) instead of j4/(8 + j2) (the current into a branch is set by the **other** branch's impedance); in case (ii) it expands (8 − j2)(3 + j2) + j12 as 12 + j11 instead of 28 + j22. The answer above was confirmed by a full node solve.

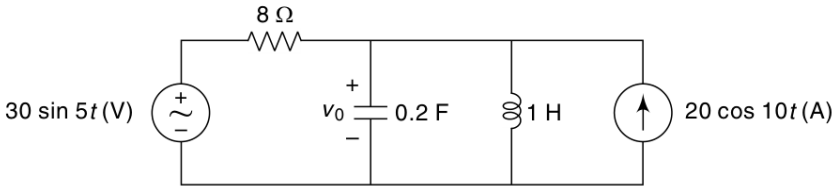
---

## A6 · Problem 5.6

Superposition · two frequencies **new idea**

**QUESTION**

Find $v_0$ using the superposition theorem.

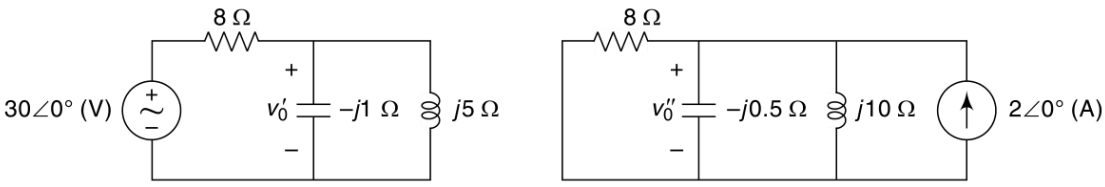



**TRY IT FIRST**

Work out $X_C$ and $X_L$ separately for ω = 5 and ω = 10 before drawing anything.

**HINT: WHAT'S NEW**

- **Reactances change with frequency.** $X_C = \frac{1}{\omega C}$ and $X_L = \omega L$ must be recomputed for each source's own ω.
- **Mixed sin and cos:** keep each term in the function it started with (sin for the voltage source, cos for the current source).




(i) voltage source alone, ω = 5 · (ii) current source alone, ω = 10

**Given:** 30 sin 5t V with 8 Ω; 0.2 F, 1 H, 20 cos 10t A. **To find:** $v_0(t)$.

**30 sin 5t alone** (ω = 5; current source opened): $X_C = \frac{1}{5 \times 0.2} = 1\ \Omega \to -j1\ \Omega$; $X_L = 5 \times 1 = 5\ \Omega \to j5\ \Omega$.

$$\frac{30 - v'}{8} = \frac{v'}{-j1} + \frac{v'}{j5} \quad \Rightarrow \quad v' = \frac{30}{8(0.125 + j0.8)} = 4.631\angle -81.12°\ \text{V}$$

**20 cos 10t alone** (ω = 10; voltage source shorted): $X_C = \frac{1}{10 \times 0.2} = 0.5\ \Omega \to -j0.5\ \Omega$; $X_L = 10\ \Omega \to j10\ \Omega$.

$$20 = v''\left(\frac{1}{8} + \frac{1}{-j0.5} + \frac{1}{j10}\right) = v''(0.125 + j1.9) \quad \Rightarrow \quad v'' = 10.50\angle -86.24°\ \text{V}$$

**ANSWER**

**$v_0(t) = 4.631 \sin(5t - 81.12°) + 10.50 \cos(10t - 86.24°)$ V**

**BOOK INCONSISTENCY IN 5.6**

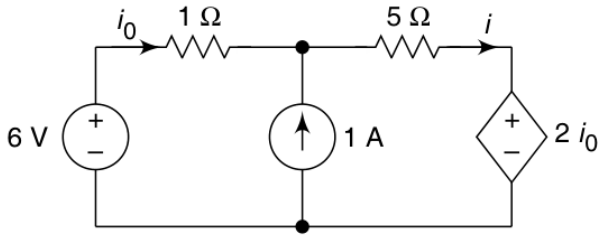
The figure gives the current source as **20** cos 10t A, but the book's solution uses **2**∠0° A and prints 1.051 cos(10t − 86.24°) for the second term. The answer above follows the figure. If your source is 2 A, the second term is 1.051 cos(10t − 86.24°) V; the first term is unchanged.

## A7 · Problem 5.7

Superposition · current-controlled source **new idea**

**QUESTION**

Find $i_0$ and i from the circuit of the figure using the superposition theorem.



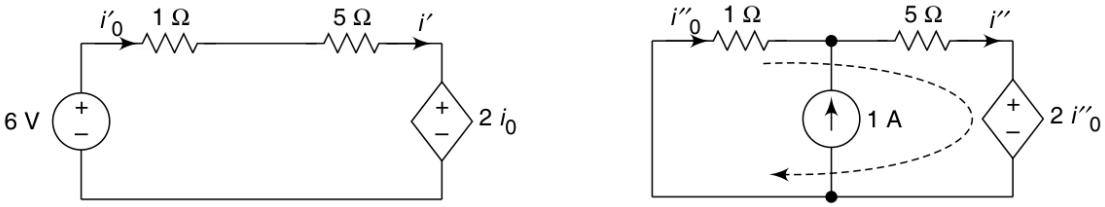

$2i_0$ is a current-controlled voltage source.

**TRY IT FIRST**

In each case, the dependent source's value is 2 × (that case's $i_0$). Write KCL at the middle node and one KVL around the outer loop.

**HINT: WHAT'S NEW**

- **Controlled by a current, not a voltage** (5.3 was voltage-controlled). The rule is the same: keep the source, and re-find its controlling current $i_0$ in each case.




(i) 6 V alone · (ii) 1 A alone

**Given:** 6 V, 1 A, 1 Ω, 5 Ω, CCVS $2i_0$. **To find:** $i_0$, i.

**6 V alone** (1 A opened): one loop, so $i_0' = i'$.

$$6 = 1\cdot i' + 5\cdot i' + 2i_0' = 8i' \quad \Rightarrow \quad i' = i_0' = 0.75 \text{ A}$$

**1 A alone** (6 V shorted):

$$\text{KCL:} \quad i'' = 1 + i_0'' \qquad \text{outer KVL:} \quad 1\cdot i_0'' + 5i'' + 2i_0'' = 0$$

$$\Rightarrow \quad 3i_0'' + 5(1 + i_0'') = 0$$

$$8i_0'' = -5 \quad \Rightarrow \quad i_0'' = -0.625 \text{ A}, \quad i'' = 0.375 \text{ A}$$

**ANSWER**

**$i = 0.75 + 0.375 = 1.125$ A, $i_0 = 0.75 - 0.625 = 0.125$ A**

**BOOK ERROR IN 5.7**

The book sets up the same equation, then solves $8i_0'' = -5$ as $i_0'' = -5/4 = -1.25$ A, and prints $i = 0.5$ A, $i_0 = -0.5$ A. The correct value is $-5/8 = -0.625$ A. Direct check: KCL $i = i_0 + 1$ and KVL $6 - i_0 - 5i = 2i_0$ give $i_0 = 1/8$ A ✓.

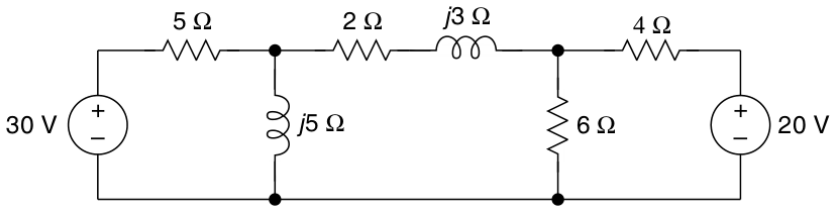
---

## A8 · Problem 5.8

Superposition · AC **practice**

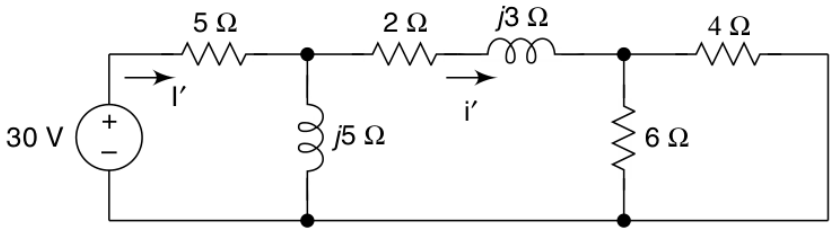
**QUESTION**

Using the superposition theorem, calculate the current through the (2 + j3) Ω impedance branch.




**TRY IT FIRST**

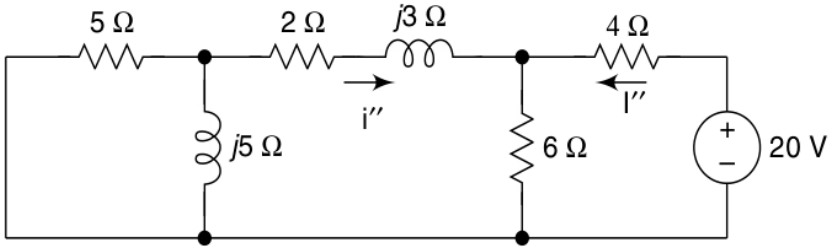
Series-parallel reduction, twice. Watch which way the branch current flows in case II.




Case I: 30 V alone

**Case I: 30 V alone** (20 V shorted). Branch: $Z_b = (2 + j3) + (6 \parallel 4) = 4.4 + j3$ Ω.

$$Z = 5 + \frac{Z_b \times j5}{Z_b + j5} = 6.32 + j2.60\ \Omega, \qquad I' = \frac{30}{Z} = 4.06 - j1.67\ \text{A}, \qquad i' = I' \times \frac{j5}{Z_b + j5} = 2.389 + j0.270\ \text{A}$$




Case II: 20 V alone

**Case II: 20 V alone** (30 V shorted). Branch: $Z_a = (5 \parallel j5) + 2 + j3 = 4.5 + j5.5$ Ω.

$$Z = 4 + \frac{6Z_a}{Z_a + 6} = 7.31 + j1.41\ \Omega, \qquad I'' = \frac{20}{Z} = 2.638 - j0.509\ \text{A}$$

$$i'' = -I'' \times \frac{6}{Z_a + 6} = -1.063 + j0.848\ \text{A} \qquad \text{(it flows right} \rightarrow \text{left, against i)}$$

**ANSWER**

**i = i′ + i″ = 1.326 + j1.117 = 1.73∠40.13° A** (= book)

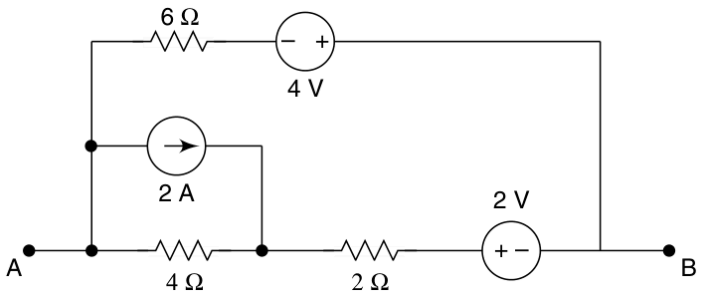
---

## A9 · Problem 5.9

Superposition · three sources, open terminals **new idea**

**QUESTION**

Using the superposition theorem, find $V_{AB}$.

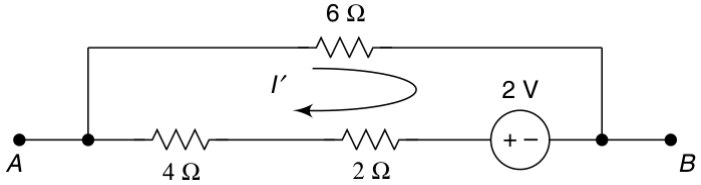



**TRY IT FIRST**

Three sources, so three cases. Nothing is connected across A–B, so no current leaves at A or B.

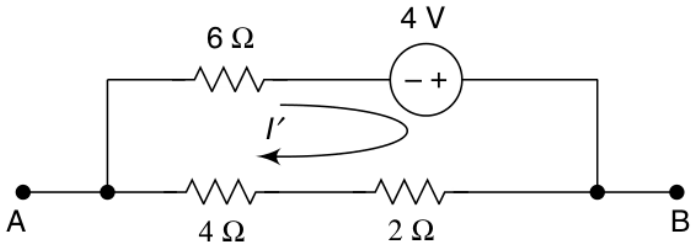
**HINT: WHAT'S NEW**

- **Three sources:** same method, one more case.
- **Open terminals:** $V_{AB}$ is the voltage across a gap. With no load, the only path is the loop 4 Ω + 2 Ω + 6 Ω, and $V_{AB}$ is the drop across the 6 Ω.
- **Source transformation inside a case:** 2 A in parallel with 4 Ω equals an 8 V source in series with 4 Ω, which makes case III a single loop.



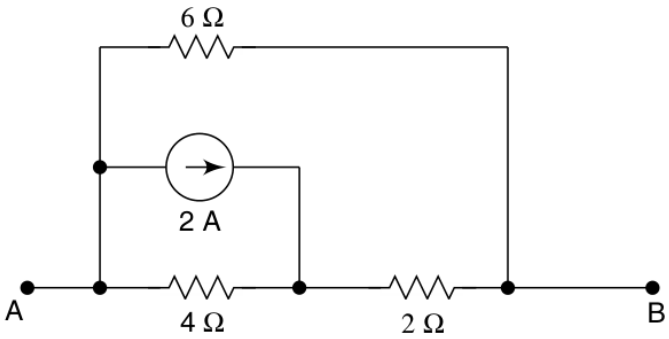

Case I: 2 V alone

Case I (2 V): $I' = \dfrac{2}{4+2+6} = \dfrac{1}{6}$ A, $V_{AB}' = 6I' = 1$ V

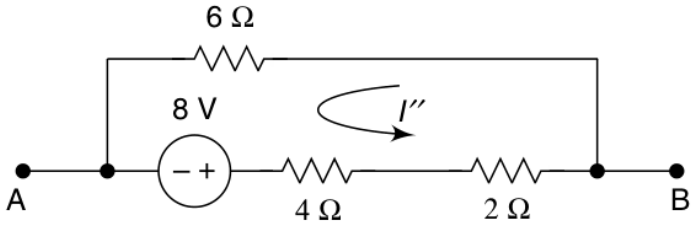



Case II: 4 V alone

Case II (4 V): $I'' = \dfrac{4}{12} = \dfrac{1}{3}$ A, $V_{AB}'' = -6I'' = -2$ V




Case III: 2 A alone




Case III after source transformation: 2 A ∥ 4 Ω → 8 V in series with 4 Ω

Case III (2 A): $I''' = \dfrac{8}{12} = \dfrac{2}{3}$ A, $V_{AB}''' = -6I''' = -4$ V

(In case I the current runs through the 6 Ω from A's side to B's side, so A is higher; in cases II and III it runs the other way, hence the minus signs.)

**ANSWER**

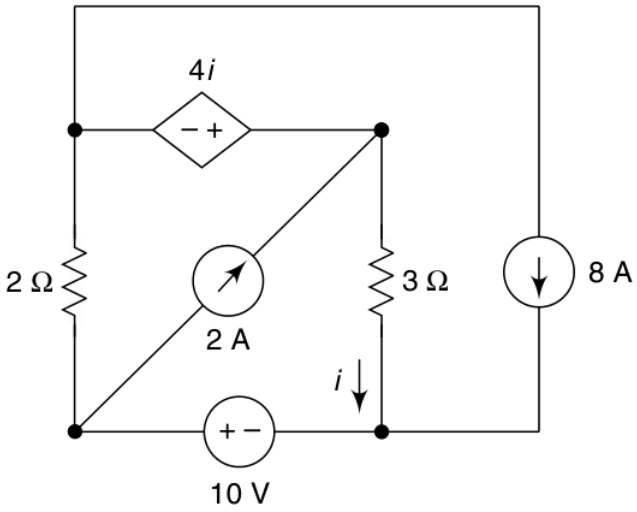
**$V_{AB} = 1 - 2 - 4 = -5$ V** (= book)

## A10 · Problem 5.10

Superposition · dependent source, three sources **new idea**

**QUESTION**

Find the current i in the circuit shown using the superposition theorem.



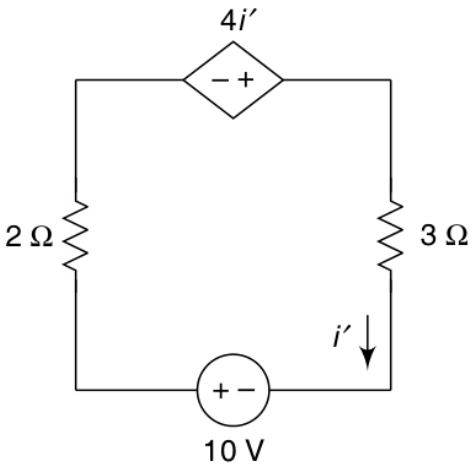

4i is a current-controlled voltage source; i flows down the 3 Ω.

**TRY IT FIRST**

Case I is one loop. In case II, look for a resistor in series with the dependent source.

**HINT: WHAT'S NEW**

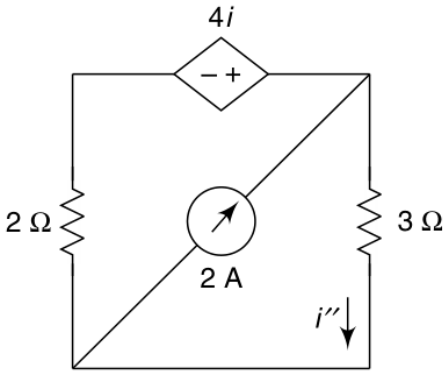
- **Source transformation works on dependent sources too.** In case II, the 4i source in series with the 2 Ω becomes a dependent current source 4i/2 = **2i** in parallel with 2 Ω.
- **A current source inside a mesh fixes that mesh current** (case III: the 8 A source sets the mesh difference I − 8).



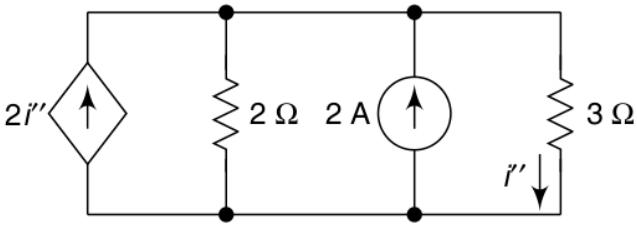

Case I: 10 V alone

**Case I: 10 V alone** (2 A and 8 A opened). Loop current i′ clockwise:

$$-4i' + 3i' - 10 + 2i' = 0 \qquad \Rightarrow \qquad i' = 10\ \text{A}$$



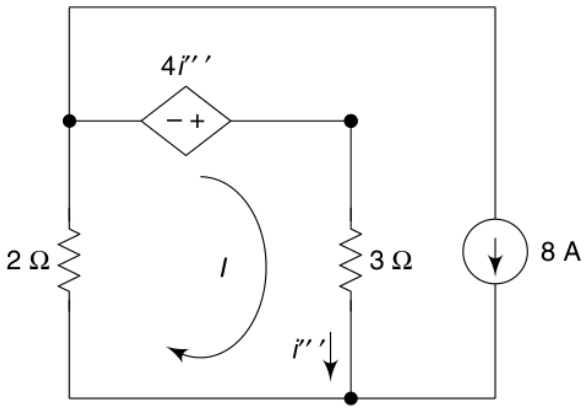

Case II: 2 A alone




Case II after converting 4i″ + 2 Ω into 2i″ ‖ 2 Ω

**Case II: 2 A alone** (10 V shorted, 8 A opened). Total (2 + 2i″) splits between 2 Ω and 3 Ω:

$$i'' = \frac{2}{2+3}(2 + 2i'') \qquad \Rightarrow \qquad 5i'' = 4 + 4i'' \qquad \Rightarrow \qquad i'' = 4\ \text{A}$$




Case III: 8 A alone

**Case III: 8 A alone** (10 V shorted, 2 A opened). Mesh current I; the 3 Ω carries i‴ = I − 8:

$$-4i''' + 3(I - 8) + 2I = 0, \qquad I = i''' + 8 \qquad \Rightarrow \qquad -4i''' + 3i''' + 2i''' + 16 = 0$$
$$\Rightarrow \qquad i''' = -16\ \text{A}$$

**ANSWER**

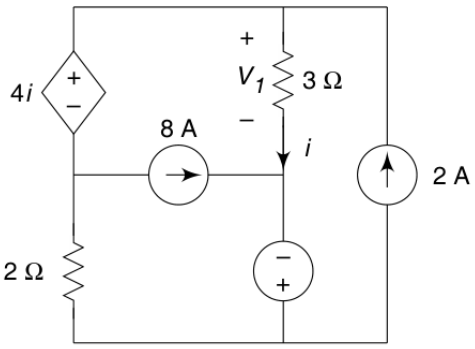
**i = 10 + 4 − 16 = −2 A** (= book)

## A11 · Problem 5.11

Superposition · dependent source, three sources **practice**

**QUESTION**

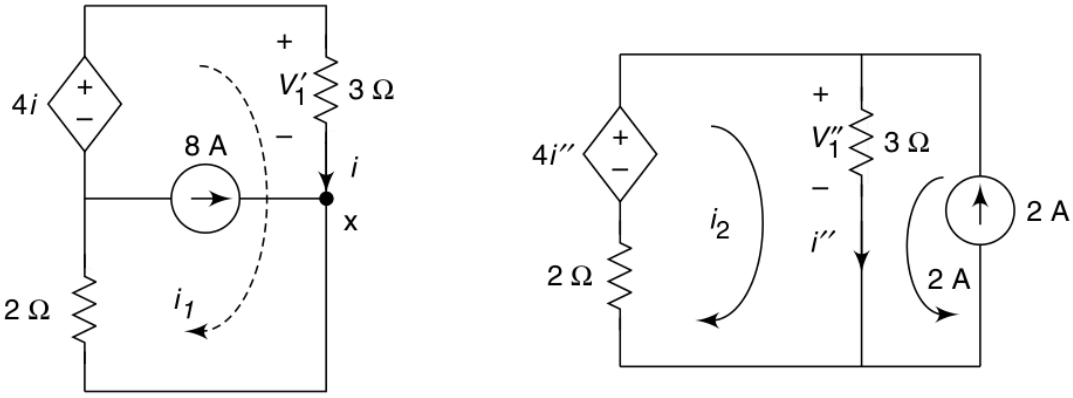
Using the superposition theorem, determine $V_1$, the voltage across the 3 Ω resistor.




The figure prints no value on the bottom-middle voltage source; the book's solution uses 10 V.

**TRY IT FIRST**

Three cases again; the dependent source 4i stays in all three. Case I has a supermesh around the 8 A source.



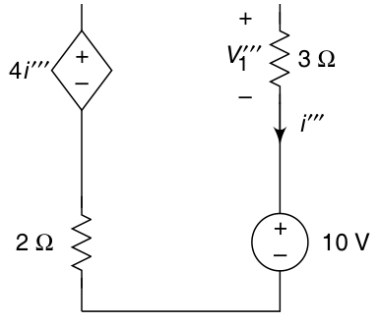

Case I: 8 A alone · Case II: 2 A alone

**Case I: 8 A alone.** Supermesh KVL and KCL at node x:

$$3i' + 2i_1 - 4i' = 0 \Rightarrow i_1 = \frac{i'}{2}; \qquad i_1 = 8 + i' \Rightarrow \frac{i'}{2} = 8 + i' \Rightarrow i' = -16 \text{ A}, \qquad V_1' = 3i' = -48 \text{ V}$$

**Case II: 2 A alone.** Mesh current $i_2$ on the left; the 3 Ω carries $i'' = i_2 + 2$:

$$3(i_2 + 2) + 2i_2 - 4i'' = 0 \Rightarrow 5i_2 + 6 - 4(i_2 + 2) = 0 \Rightarrow i_2 = 2 \text{ A}, \qquad i'' = 4 \text{ A}, \qquad V_1'' = 12 \text{ V}$$




Case III: 10 V alone

**Case III: 10 V alone.** One loop:

$$3i''' - 10 + 2i''' - 4i''' = 0 \Rightarrow i''' = 10\ \text{A}, \qquad V_1''' = 30\ \text{V}$$

**ANSWER**

**$V_1 = -48 + 12 + 30 = -6$ V** (= book)

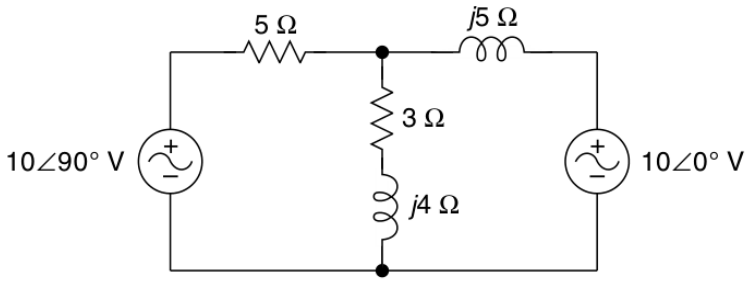
---

## A12 · Problem 5.12

Superposition · AC **new idea**

**QUESTION**

Calculate the current through the impedance (3 + j4) Ω using the superposition theorem.

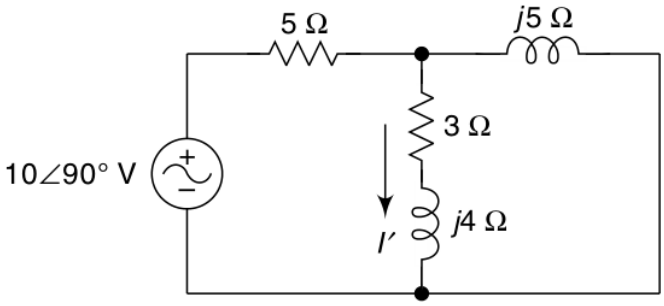



**TRY IT FIRST**

Two voltage sources, same frequency. Get each case's current as a fraction before simplifying.

**HINT: WHAT'S NEW**

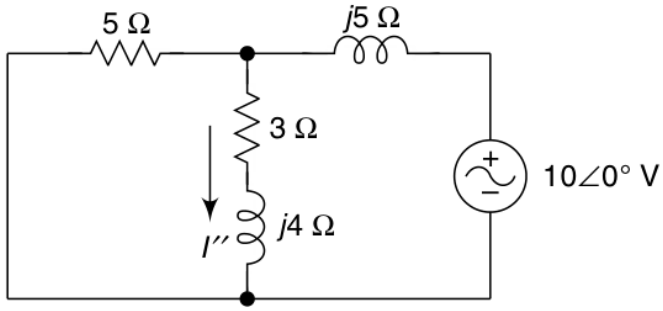
- **The two contributions can cancel.** Here they come out equal and opposite, so the branch current is zero. A zero answer is not an error; check it with one node equation.




10∠90° V alone

10∠90° alone: $I = \dfrac{j10}{5 + \dfrac{(3 + j4)j5}{3 + j9}} = \dfrac{j10(3 + j9)}{-5 + j60}$, $I' = I \times \dfrac{j5}{3 + j9} = \dfrac{-10}{-1 + j12} =$

$0.069 + j0.828$ A




10∠0° V alone

10∠0° alone: $I = \dfrac{10}{j5 + \dfrac{5(3 + j4)}{8 + j4}} = \dfrac{10(8 + j4)}{-5 + j60}$, $I'' = I \times \dfrac{5}{8 + j4} = \dfrac{10}{-1 + j12} =$

$-0.069 - j0.828$ A

**ANSWER**

**I = I′ + I″ = 0 A** (= book)

**CHECK**

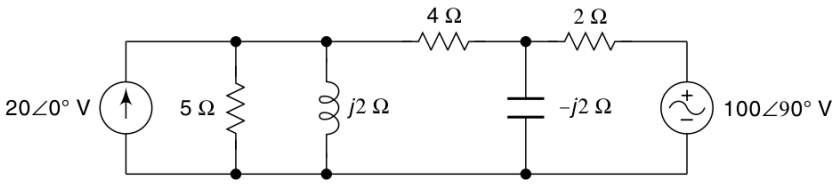
Node equation at the top node: $V(\frac{1}{5} + \frac{1}{3 + j4} + \frac{1}{j5}) = \frac{j10}{5} + \frac{10}{j5} = j2 - j2 = 0$, so V = 0 and no current flows in 3 + j4 ✓.

## A13 · Problem 5.13

Superposition · AC **practice**

**QUESTION**

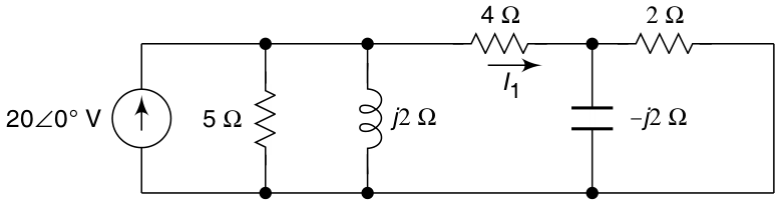
Using the superposition theorem, determine the current in the 4 Ω resistor.




The book labels the current source "20∠0° V"; it is a current source (arrow), 20∠0° A.

**TRY IT FIRST**

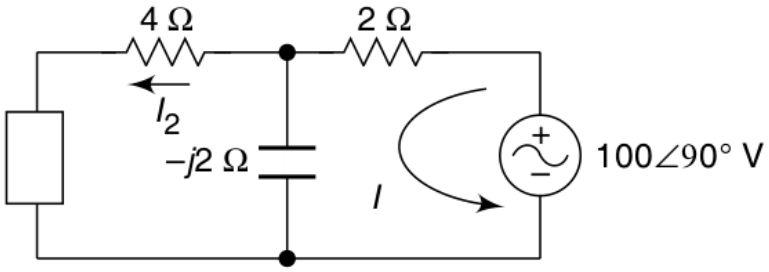
Combine the parallel pairs first: 5 Ω ∥ j2 Ω and 2 Ω ∥ −j2 Ω.




Case I: 20∠0° A alone

**Case I: 20∠0° A alone** (voltage source shorted). $Z_1 = 5 \parallel j2 = 0.690 + j1.724\ \Omega$; $Z_2 = 2 \parallel (-j2) = 1 - j1\ \Omega$.

$$I_1 = 20 \times \frac{Z_1}{Z_1 + 4 + Z_2} = 3.145 + j5.660\ \text{A} \qquad \text{(left} \rightarrow \text{right)}$$




Case II: 100∠90° V alone

**Case II: 100∠90° V alone** (current source opened). $(Z_1 + 4) \parallel (-j2) = 0.850 - j1.950\ \Omega$:

$$I_s = \frac{j100}{2 + (0.850 - j1.950)} = -16.35 + j23.90\ \text{A}, \qquad V_B = I_s(0.850 - j1.950) = 32.70 + j52.20\ \text{V}$$

$$I_2 = \frac{V_B}{Z_1 + 4} = 9.748 + j7.547\ \text{A} \qquad \text{(right} \rightarrow \text{left)}$$

**ANSWER**

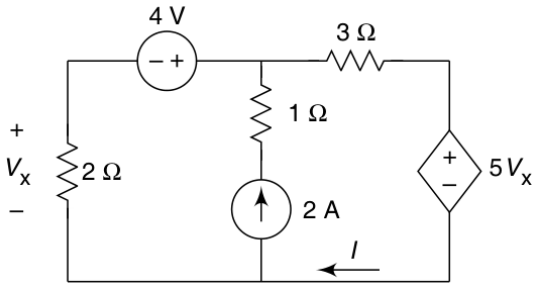
**$I = I_1 - I_2 = -6.604 - j1.887 = 6.87\angle -164.05°$ A (left → right)**, i.e. 6.87 A flowing right → left (= book, 6.89∠−163.67° A, to rounding)

## A14 · Problem 5.14

Superposition · dependent source **practice**

**QUESTION**

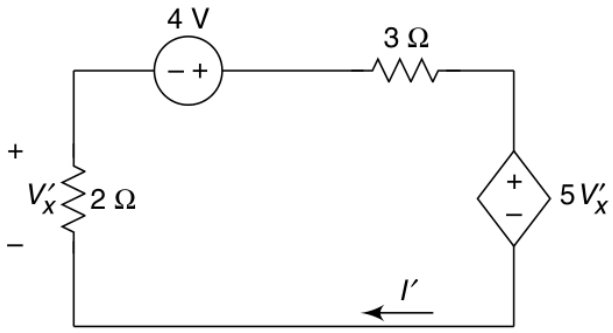
Find I in the figure using the superposition theorem.




$5V_x$ is a voltage-controlled voltage source; $V_x$ is across the 2 Ω (+ at top).

**TRY IT FIRST**

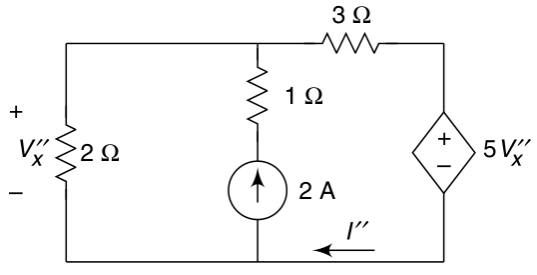
Same idea as 5.3. In case II, the 4 V short joins the two top-left nodes; write one KCL there.




Case I: 4 V alone

**Case I: 4 V alone** (2 A opened). One loop; I′ flows up the 2 Ω, so $V_x' = -2I'$:

$$-4 + 3I' + 5V_x' - V_x' = 0 \Rightarrow 3I' + 4(-2I') = 4 \Rightarrow I' = -0.8 \text{ A}$$




Case II: 2 A alone

**Case II: 2 A alone** (4 V shorted). KCL at the top node (voltage $V_x''$):

$$2 = \frac{V_x''}{2} + \frac{V_x'' - 5V_x''}{3} \Rightarrow V_x'' = -2.4 \text{ V}, \qquad I'' = \frac{V_x'' - 5V_x''}{3} = 3.2 \text{ A}$$

**ANSWER**

**I = −0.8 + 3.2 = 2.4 A** (= book)

## B · Thevenin's and Norton's theorems

Superposition: one source at a time (done) → **Thevenin: whole network → one source + one impedance** → Norton: the same as a current source → Maximum power transfer (next)

**Where this sits.** Superposition showed that a linear circuit's response is a sum of simple parts. Thevenin's theorem uses that fact to shrink everything a load is connected to into **two numbers**, so changing the load never means re-solving the whole circuit.

**Three terms first.**

- **Terminals a–b:** the two points where the load connects. Everything on the other side of them is the **network**.
- **Open-circuit voltage $V_{oc}$:** the voltage across a–b with the load removed (nothing connected).
- **Short-circuit current $I_{sc}$:** the current through a wire placed across a–b, counted **from a to b through the wire** (out of the + terminal).

**Statement.** Seen from a–b, any linear network behaves exactly like **one voltage source $V_{th} = V_{oc}$ in series with one impedance $Z_{th}$**. Norton's form is the same network as **one current source $I_N = I_{sc}$ in parallel with $Z_N = Z_{th}$**. The two forms are a source transformation of each other: $V_{th} = I_N Z_{th}$.

**Finding $Z_{th}$: three cases.**

| The network contains | $Z_{th}$ is found by |
|---|---|
| Independent sources only | switch every source off (V → short, I → open), then combine impedances seen from a–b |
| Independent **and** dependent sources | $Z_{th} = V_{oc} / I_{sc}$, or switch off the independent sources only and apply a test source |
| Dependent sources only | $V_{th} = 0$; apply a test source (1 A in, measure V): $Z_{th} = V / I$ |

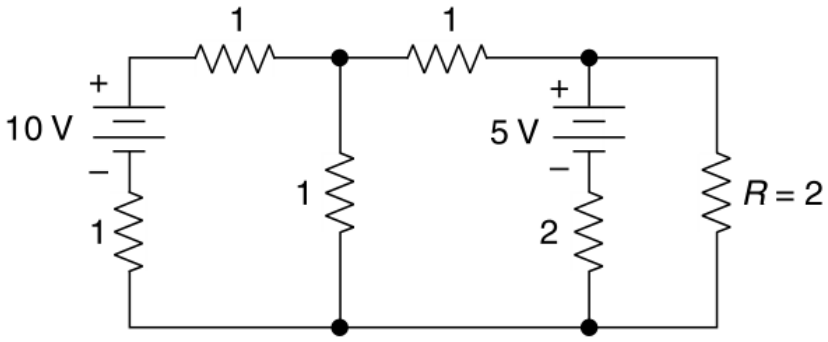
With dependent sources $Z_{th}$ can come out **negative** (the network supplies energy). Keep the sign of $I_{sc}$: it decides the sign of $Z_{th}$.

---

### A15 · Problem 5.15

Thevenin · DC **teach**

> **QUESTION**
> Draw the Thevenin's equivalent of the circuit in the figure and find the load current i. All values are in ohm.



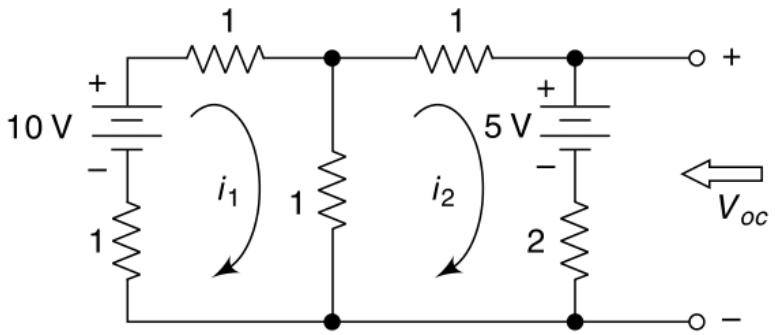

i flows down through R = 2 Ω.

**TRY IT FIRST**

Remove R. Find the voltage across its terminals (two meshes), then short both sources and find the resistance seen from there. Last, put R back on the equivalent.

**HINT: WHAT'S NEW**

- **The Thevenin recipe:** (1) remove the load; (2) $V_{th}$ = open-circuit voltage at its terminals; (3) $R_{th}$ = resistance seen into the terminals with every independent source switched off; (4) load current = $V_{th} / (R_{th} + R_L)$.
- **Switching off for $R_{th}$** uses the same rule as superposition: voltage sources become shorts.




Load removed: two mesh currents, $V_{oc}$ across the open terminals

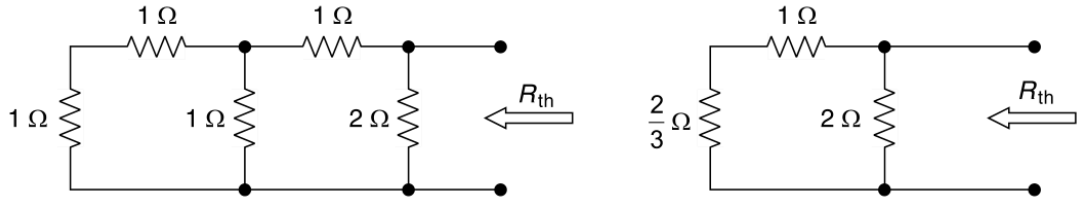
**Given:** 10 V, 5 V; 1 Ω (series with 10 V), 1 Ω, 1 Ω, 2 Ω; R = 2 Ω. **To find:** $V_{th}$, $R_{th}$, i.

**Step 1: $V_{th}$** (R removed). Mesh currents clockwise:

$$\text{Mesh 1:} \quad 3i_1 - i_2 = 10 \qquad \text{Mesh 2:} \quad -i_1 + 4i_2 = -5$$

$$\Rightarrow \quad i_1 = \frac{35}{11}\ \text{A}, \qquad i_2 = -\frac{5}{11}\ \text{A}$$

$$V_{th} = V_{oc} = 5 + 2i_2 = 5 - \frac{10}{11} = \frac{45}{11} = 4.091\ \text{V}$$




Both sources shorted: $R_{th}$ seen from the load terminals

**Step 2: $R_{th}$** (both sources shorted):

$$(1 + 1) \parallel 1 = \frac{2}{3}\ \Omega, \qquad \frac{2}{3} + 1 = \frac{5}{3}\ \Omega, \qquad R_{th} = \frac{5}{3} \parallel 2 = \frac{10}{11} = 0.909\ \Omega$$

**Step 3: load current:**

$$i = \frac{V_{th}}{R_{th} + R} = \frac{45/11}{10/11 + 2} = \frac{45}{32}\ A$$

**ANSWER**

**Thevenin equivalent: 45/11 V in series with 10/11 Ω. i = 1.406 A** (= book)

**SIDE NOTE**

The book prints $i_1 = 5/11$; it is 35/11 (only $i_2$ is used, so the answer is unaffected). Full node solve with R in place: i = 1.40625 A ✓.

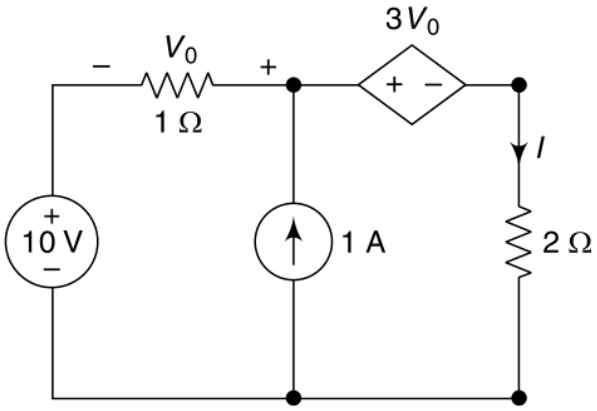
---

## A16 · Problem 5.16

Thevenin · dependent source **teach**

**QUESTION**

Find I in the given figure, using Thevenin's theorem.



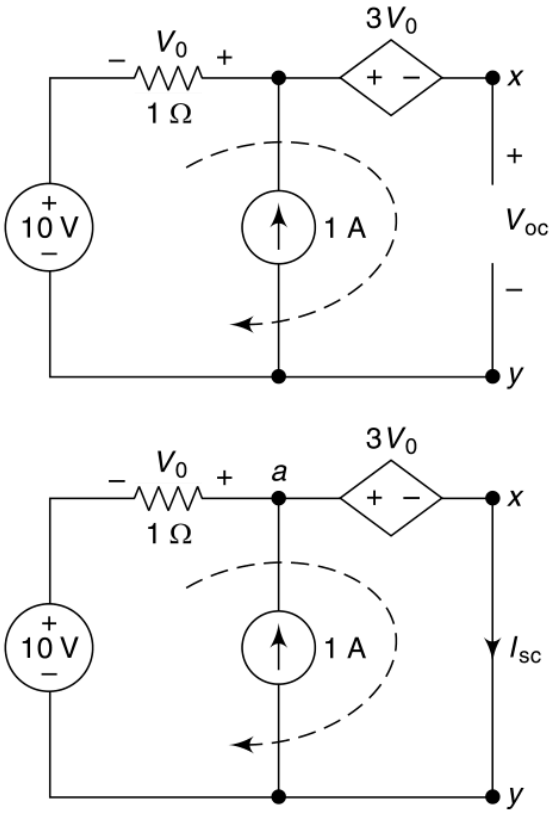

$3V_0$ is a voltage-controlled voltage source; $V_0$ is across the 1 Ω (+ on the right).

**TRY IT FIRST**

The dependent source cannot be switched off, so $R_{th}$ must come from $V_{oc}$ / $I_{sc}$. Find both, and keep the sign of $I_{sc}$.

**HINT: WHAT'S NEW**

- **Dependent source ⇒ $R_{th} = V_{oc} / I_{sc}$** (or a test source). Shorting the independent sources and combining resistors would ignore the dependent source.
- **$I_{sc}$ is measured from x to y through the short.** A negative $I_{sc}$ gives a **negative $R_{th}$**, which is allowed: the dependent source can supply energy.




Top: terminals x–y open ($V_{oc}$). Bottom: x–y shorted ($I_{sc}$).

**Given:** 10 V, 1 A, 1 Ω, VCVS $3V_0$, load 2 Ω. **To find:** I.

**$V_{oc}$** (2 Ω removed). Nothing leaves at x, so the 1 A must flow left through the 1 Ω: $V_0 = 1 \times 1 = 1$ V.

$$V_a = 10 + V_0 = 11 \text{ V}, \qquad V_{oc} = V_a - 3V_0 = 11 - 3 = 8 \text{ V}$$

**$I_{sc}$** (x shorted to y, so $V_x = 0$):

$$0 = V_a - 3V_0 = (10 + V_0) - 3V_0 \qquad \Rightarrow \qquad V_0 = 5 \text{ V}$$

$$\text{KCL at a:} \qquad 1 = \frac{V_0}{1} + I_{sc} \qquad \Rightarrow \qquad I_{sc} = 1 - 5 = -4\ \text{A}$$

$$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{8}{-4} = -2\ \Omega$$

**Load:** $R_{th} + 2\ \Omega = -2 + 2 = 0$, so

$$I = \frac{V_{th}}{R_{th} + 2} = \frac{8}{0} \qquad \Rightarrow \qquad \text{no finite solution}$$

**ANSWER**

**$V_{th} = 8$ V, $R_{th} = -2\ \Omega$. With the 2 Ω load, $R_{th} + R_L = 0$: the circuit as drawn has no solution.** (Book: 2 A.)

**BOOK ERROR IN 5.16**

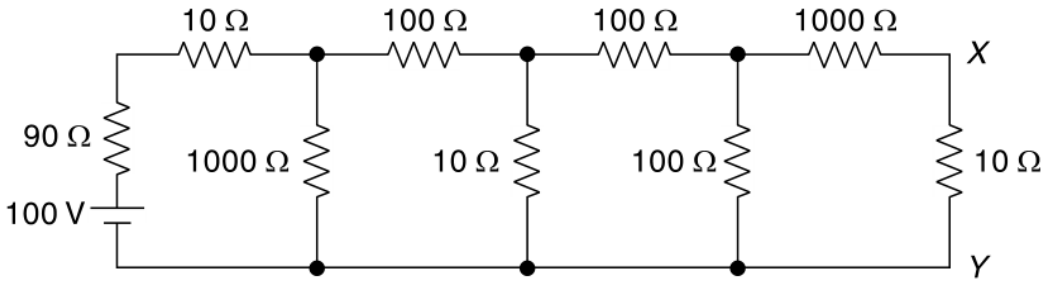
The book finds $I_{sc} = -4$ A, then writes $R_{th} = 8/4 = 2\ \Omega$, dropping the minus sign, and prints $I = 2$ A. Direct check with the 2 Ω in place: KCL at a gives $V_0 = 1 - I$; the right-hand path gives $V_x = 10 - 2V_0 = 8 + 2I$; but $V_x = 2I$, so $8 = 0$, a contradiction. A test source (10 V shorted, 1 A opened, 1 A pushed into x) confirms $R_{th} = -2\ \Omega$. For any other load, $I = \dfrac{8}{R_L - 2}$.

## A17 · Problem 5.17

Thevenin · repeated reduction **new idea**

**QUESTION**

By the iterative use of Thevenin's theorem, reduce the circuit shown in the figure to a single emf acting in series with a single resistor. Hence, calculate the current in the 10 Ω resistor connected across XY.




Ladder network; the load is the 10 Ω across XY.

**TRY IT FIRST**

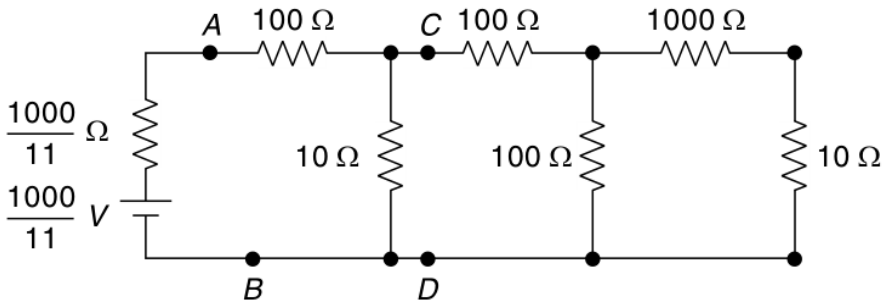
Cut the ladder just right of the first shunt resistor and replace the left part by its Thevenin equivalent. Repeat, one rung at a time.

**HINT: WHAT'S NEW**

- **Iterative Thevenin:** a ladder is reduced rung by rung. Each step is a voltage divider for $V_{th}$ and a parallel combination for $R_{th}$: $V_{new} = V_{old} \times \dfrac{R_{sh}}{R_{old} + R_{series} + R_{sh}}$, $R_{new} = (R_{old} + R_{series}) \parallel R_{sh}$.

**Step 1** (left of AB: 100 V behind 90 + 10 = 100 Ω, shunt 1000 Ω):

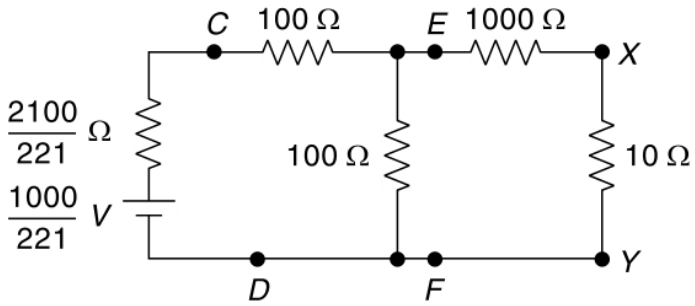
$$V_1 = 100 \times \frac{1000}{1100} = \frac{1000}{11}\ \text{V}, \qquad R_1 = 100 \parallel 1000 = \frac{1000}{11}\ \Omega$$




After step 1

**Step 2** (add 100 Ω series, shunt 10 Ω):

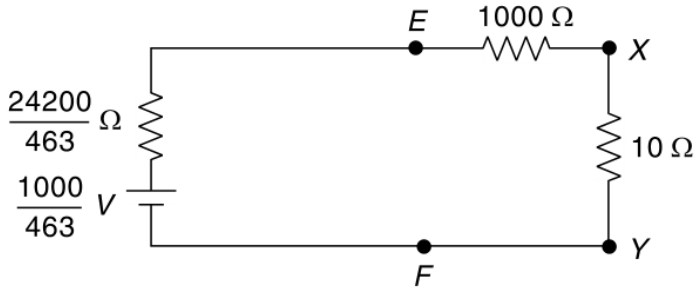
$$V_2 = \frac{1000}{11} \times \frac{10}{2100/11 + 10} = \frac{1000}{221}\ \text{V}, \qquad R_2 = \frac{2100}{11} \parallel 10 = \frac{2100}{221}\ \Omega$$




After step 2

**Step 3** (add 100 Ω series, shunt 100 Ω):

$$V_3 = \frac{1000}{221} \times \frac{100}{24200/221 + 100} = \frac{1000}{463}\ \text{V}, \qquad R_3 = \frac{24200}{221} \parallel 100 = \frac{24200}{463}\ \Omega$$




After step 3, with the 1000 Ω still in series

**Step 4** (add 1000 Ω series):

$$V_{th} = \frac{1000}{463} = 2.160\ \text{V}, \qquad R_{th} = \frac{24200}{463} + 1000 = \frac{487200}{463} = 1052.3\ \Omega$$

$$I = \frac{1000/463}{487200/463 + 10} = \frac{1000}{491830}\ \text{A}$$

**ANSWER**

**Single emf 2.160 V in series with 1052.3 Ω; I = 2.03 mA ≈ 0.002 A** (= book)

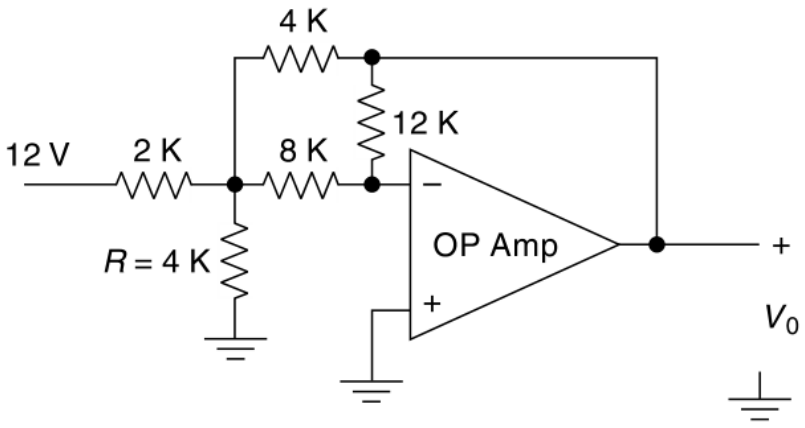
---

## A18 · Problem 5.18

Thevenin · op-amp, test source **new idea**

**QUESTION**

In the operational-amplifier circuit shown in the figure, find I in the R = 4 kΩ resistor, using Thevenin's theorem.




Ideal op-amp; the load is R = 4 kΩ from node $e_1$ to ground.

**TRY IT FIRST**

With R removed, write KCL at $e_1$ and at the inverting input. For $R_{th}$, short the 12 V and push 1 A into $e_1$.

**HINT: WHAT'S NEW**

- **Ideal op-amp rules:** no current into its inputs, and (with negative feedback) $e_2 = e_+ = 0$ V. The output can supply any current, so **no KCL is written at the output node**.
- **Test source for $R_{th}$:** the op-amp acts like a dependent source, so switch off the 12 V only, inject 1 A at the terminals and read the voltage: $R_{th} = e_1 / 1$ A.

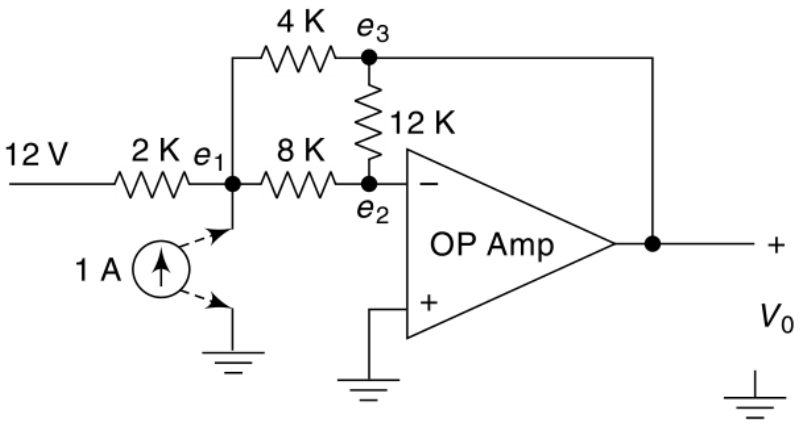
**Given:** 12 V, 2 kΩ, 4 kΩ (feedback $e_1 \to V_0$), 8 kΩ, 12 kΩ, ideal op-amp. **To find:** I in R = 4 kΩ.

**$V_{th}$** (R removed; $e_2 = 0$, $e_3 = V_0$):

$$\text{KCL at } e_2: \quad \frac{0 - e_1}{8k} + \frac{0 - V_0}{12k} = 0 \quad \Rightarrow \quad V_0 = -\frac{3}{2}e_1$$

$$\text{KCL at } e_1: \quad \frac{e_1 - 12}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 0 \quad \Rightarrow \quad 7e_1 = 48 + 2V_0$$

$$7e_1 = 48 - 3e_1 \quad \Rightarrow \quad V_{th} = e_1 = 4.8 \text{ V}$$




12 V shorted, 1 A test source at the load terminals

**$R_{th}$** (12 V shorted, 1 A injected at $e_1$):

$$\frac{e_1}{2k} + \frac{e_1 - V_0}{4k} + \frac{e_1}{8k} = 1 \quad \Rightarrow \quad 7e_1 - 2V_0 = 8000$$

$$10e_1 = 8000 \quad \Rightarrow \quad e_1 = 800 \text{ V}$$

$$R_{th} = \frac{800 \text{ V}}{1 \text{ A}} = 800 \ \Omega, \qquad I = \frac{4.8}{800 + 4000} = 1 \times 10^{-3} \text{ A}$$

**ANSWER**

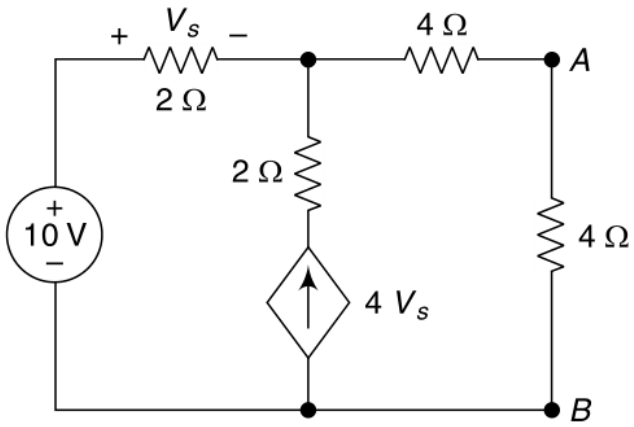
**$V_{th}$ = 4.8 V, $R_{th}$ = 800 Ω, I = 1 mA** (= book)

## A19 · Problem 5.19

Thevenin · dependent current source **practice**

**QUESTION**

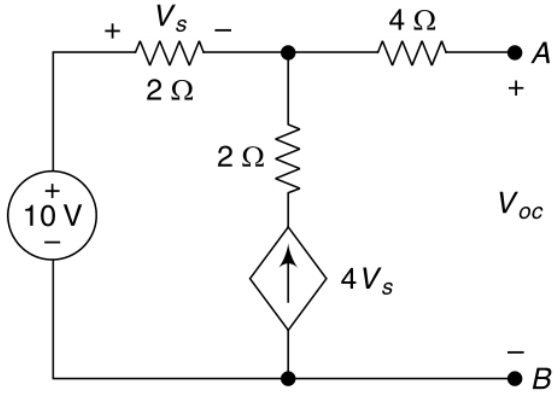
Find Thevenin's equivalent about AB for the circuit shown in the figure.




The 4 Ω across AB is the load (the book removes it).

**TRY IT FIRST**

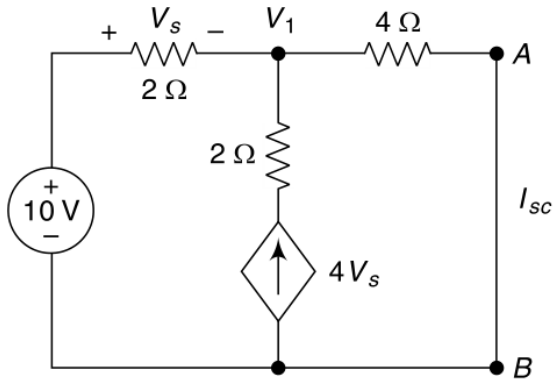
$V_{oc}$: with AB open, what does KCL at the middle node say about $V_s$? Then short AB for $I_{sc}$.




AB open

**$V_{oc}$** (AB open, so the series 4 Ω carries nothing; $V_s = 10 - V_M$). KCL at the middle node M:

$$\frac{V_s}{2} + 4V_s = 0 \quad \Rightarrow \quad V_s = 0 \quad \Rightarrow \quad V_{oc} = V_M = 10\ \text{V}$$




AB shorted

**$I_{sc}$** (AB shorted; $V_1 = V_M$, $V_s = 10 - V_1$):

$$\frac{V_1 - 10}{2} + \frac{V_1}{4} = 4(10 - V_1) \quad \Rightarrow \quad 19V_1 = 180 \quad \Rightarrow \quad V_1 = 9.474 \text{ V}$$

$$I_{sc} = \frac{V_1}{4} = 2.368 \text{ A}$$

$$R_{th} = \frac{10}{2.368} = 4.22 \ \Omega$$

**ANSWER**

**$V_{th}$ = 10 V, $R_{th}$ = 38/9 = 4.22 Ω** (= book)

**SIDE NOTE**

The 2 Ω in series with the dependent current source never matters (5.1's shortcut). Test-source check (10 V shorted, 1 A into A): $V_A = 4 + \frac{1}{4.5} = 4.22$ V ⇒ $R_{th} = 4.22$ Ω ✓.

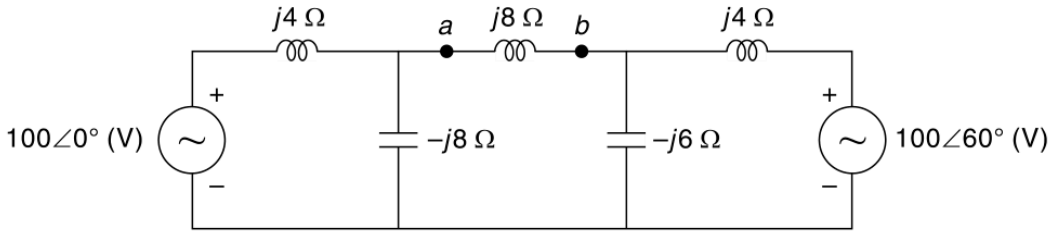
---

## A20 · Problem 5.20

Thevenin · AC, floating terminals **new idea**

**QUESTION**

In the network, determine the steady current in the 8 Ω inductor using Thevenin's theorem.



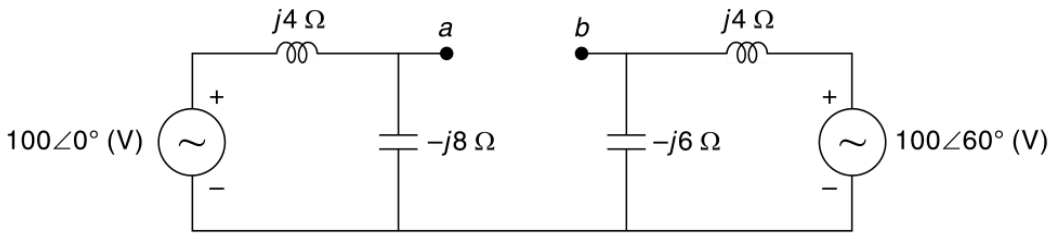

The load is the j8 Ω between a and b.

**TRY IT FIRST**

With j8 removed, the circuit splits into two separate halves. Find $V_a$ and $V_b$ with a divider each.

**HINT: WHAT'S NEW**

- **Neither terminal is ground.** $V_{th} = V_a - V_b$, each found from its own half.
- With the sources shorted, each half is a parallel pair, and the two pairs are **in series** between a and b.




j8 Ω removed: two independent halves

$$V_a = 100\angle 0° \times \frac{-j8}{j4 - j8} = 200\angle 0° \text{ V}, \qquad V_b = 100\angle 60° \times \frac{-j6}{j4 - j6} = 300\angle 60° \text{ V}$$

$$V_{th} = V_a - V_b = 200 - (150 + j259.81) = 50 - j259.81 = 264.58\angle -79.11° \text{ V}$$

$$Z_{th} = \frac{(j4)(-j8)}{j4 - j8} + \frac{(j4)(-j6)}{j4 - j6} = j8 + j12 = j20\ \Omega$$

$$i = \frac{V_{th}}{Z_{th} + j8} = \frac{264.58\angle -79.11°}{28\angle 90°}$$

**ANSWER**

**i = 9.45∠−169.11° A (a → b)** (= book)

---

## A21 · Problem 5.21

Thevenin / Norton · AC, s-domain, dependent source **new idea**

**QUESTION**

Obtain Thevenin's equivalent circuit with respect to terminals A–B in the networks shown below.

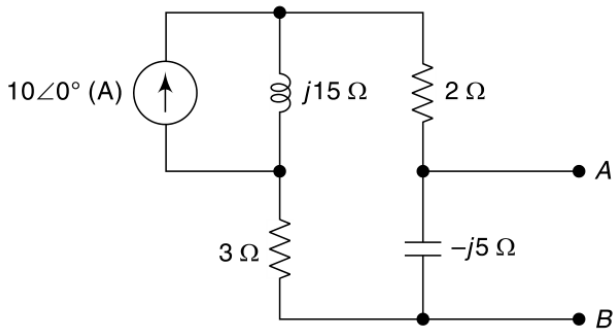
**TRY IT FIRST**

(a)–(c) are practice. In (d), treat s like jω. In (e) the dependent source stays, so use $V_{oc}$ / $I_{sc}$.

**HINT: WHAT'S NEW**

- **(d) s-domain:** impedances are R, sL and $\frac{1}{sC}$; the source becomes its Laplace transform, 2 cos 2t $\rightarrow \frac{2s}{s^2 + 4}$. Everything else is unchanged.
- **(e) AC with a dependent source:** find $V_{th}$ and $I_N$ separately; $Z_{th} = V_{th} / I_N$.

**(a)**




Source transformation: $10\angle 0°$ A ‖ j15 Ω → j150 V in series with j15 Ω. With AB open, one loop:

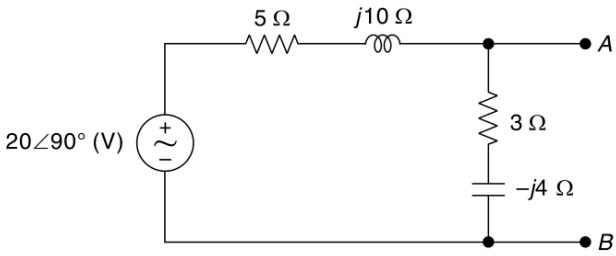
$$I = \frac{j150}{j15 + 2 - j5 + 3} = \frac{j150}{5 + j10} = 13.42\angle 26.57° \text{ A}, \qquad V_{th} = I \times (-j5) = 30 - j60 \text{ V}$$

$$Z_{th} = (-j5) \parallel (5 + j15) = \frac{-j5(5 + j15)}{5 + j10} = 1 - j7 \ \Omega$$

**ANSWER**

**(a) $V_{th} = 67.08\angle -63.43°$ V, $Z_{th} = 7.07\angle -81.87° \ \Omega$** (= book)

**(b)**



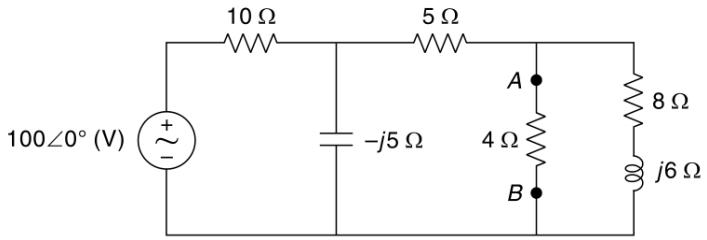

$$V_{th} = 20\angle 90° \times \frac{3 - j4}{(5 + j10) + (3 - j4)} = \frac{j20(3 - j4)}{8 + j6} = \frac{100\angle 36.87°}{10\angle 36.87°} = 10\angle 0° \text{ V}$$

$$Z_{th} = \frac{(5 + j10)(3 - j4)}{8 + j6} = \frac{55 + j10}{8 + j6} = 5 - j2.5 \ \Omega$$

**ANSWER**

**(b) $V_{th} = 10\angle 0°$ V, $Z_{th} = 5.59\angle -26.57°$ Ω** (= book; its first line prints j120, a typo for j20)

**(c)** The 4 Ω across AB is the load.




$$Z = 10 + (-j5) \parallel (13 + j6) = 11.91 - j5.15 = 12.98\angle -23.37° \ \Omega$$

$$I = \frac{100}{Z} = 7.706\angle 23.37° \text{ A}$$

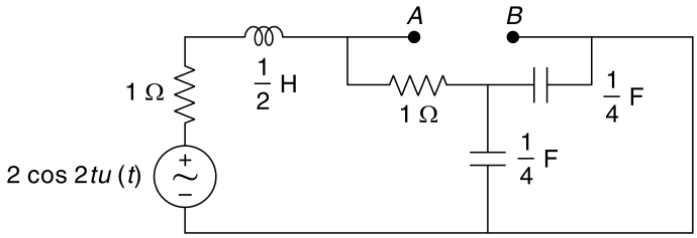
$$I_{right} = I \times \frac{-j5}{13 + j1} = 2.955\angle -71.03° \text{ A}, \qquad V_{th} = I_{right}(8 + j6) = 29.55\angle -34.16° \text{ V}$$

$$Z_{th} = [(10 \parallel -j5) + 5] \parallel (8 + j6) = (7 - j4) \parallel (8 + j6) = 5.328 - j0.044 \ \Omega$$

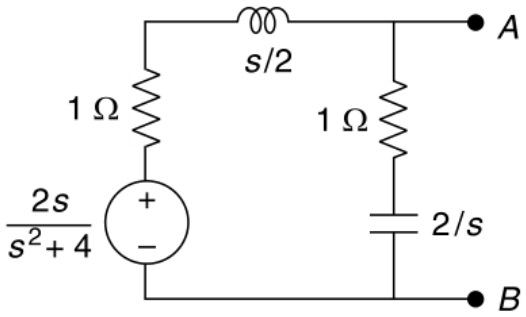
**ANSWER**

**(c) $V_{th} = 29.55\angle -34.16°$ V, $Z_{th} = 5.33\angle -0.47°$ Ω** (= book)

**(d)**




The two $\frac{1}{4}$ F capacitors both run from the same node to ground, so they add: $\frac{1}{2}$ F → $\frac{2}{s}$. A is the node after the inductor; B is ground.




Redrawn in the s-domain

$$V_{th}(s) = \frac{2s}{s^2+4} \times \frac{1 + 2/s}{(1 + 2/s) + (1 + s/2)} = \frac{2s}{s^2+4} \times \frac{2}{s+2} = \frac{4s}{(s^2+4)(s+2)}$$

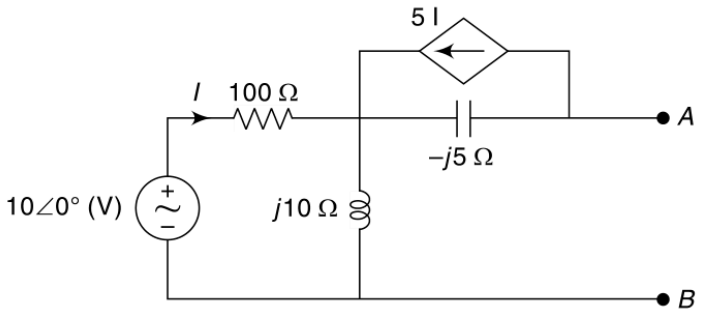
(Simplify: $1 + 2/s = \frac{s+2}{s}$ and $2 + s/2 + 2/s = \frac{(s+2)^2}{2s}$, so the ratio is $\frac{2}{s+2}$.)

$$Z_{th}(s) = (1 + 2/s) \parallel (1 + s/2) = \frac{(s+2)^2/(2s)}{(s+2)^2/(2s)} = 1\ \Omega$$

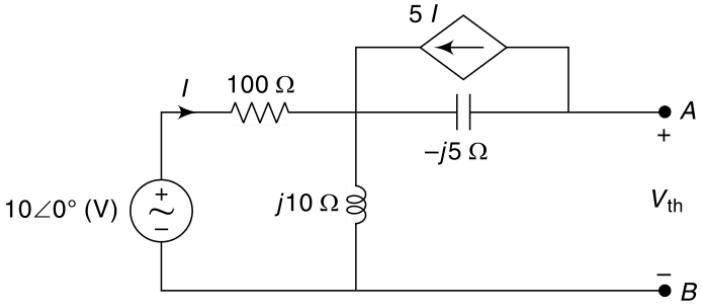
**ANSWER**

**(d) $V_{th}(s) = \frac{4s}{(s^2+4)(s+2)}$ V, $Z_{th} = 1\ \Omega$** (= book)

**(e)**




I flows through the 100 Ω; the dependent source 5I points from A towards the middle node.

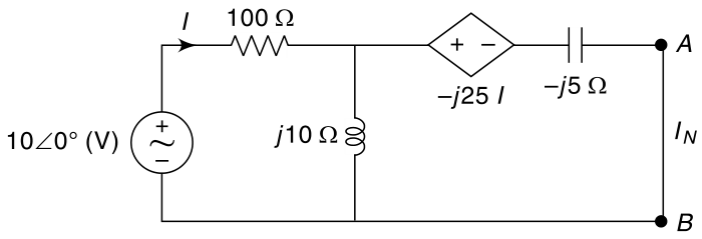



AB open

**$V_{th}$.** With AB open, the 5I circulates only through the −j5 Ω, so the source current is set by the left loop alone:

$$I = \frac{10\angle 0°}{100 + j10} = 0.0995\angle -5.71° \text{ A}$$

$$V_{th} = V_{AB} = I(j10) - 5I(-j5) = j35I = 3.483\angle 84.29° \text{ V}$$




AB shorted, with 5I ‖ −j5 transformed to −j25I in series with −j5

**$I_N$.** AB shorted. Right loop: $j10(I - I_N) + j25I + j5I_N = 0 \Rightarrow I_N = 7I$. Left loop:

$$10 = (100 + j10)I - j10I_N = (100 - j60)I \qquad \Rightarrow \qquad I = 0.0857\angle 30.96° \text{ A}$$

$$I_N = 7I = 0.600\angle 30.96° \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = \frac{3.483\angle 84.29°}{0.600\angle 30.96°} = 5.80\angle 53.33° \ \Omega$$

**ANSWER**

**(e) $V_{th} = 3.48\angle 84.3°$ V, $I_N = 0.600\angle 31.0°$ A, $Z_{th} = 5.80\angle 53.3°$ Ω** (= book)

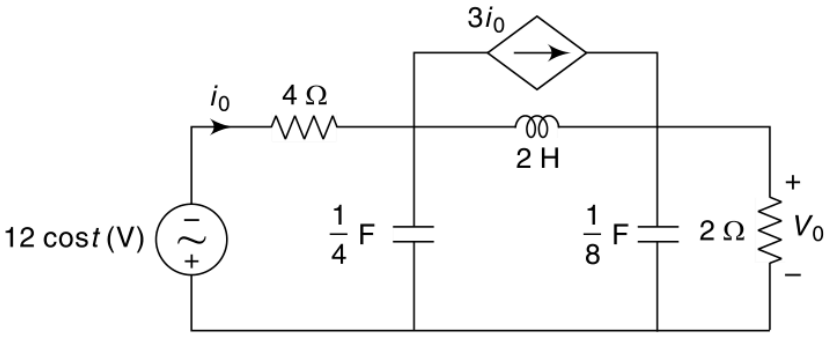
---

## A22 · Problem 5.22

Thevenin · AC, dependent source **practice**

**QUESTION**

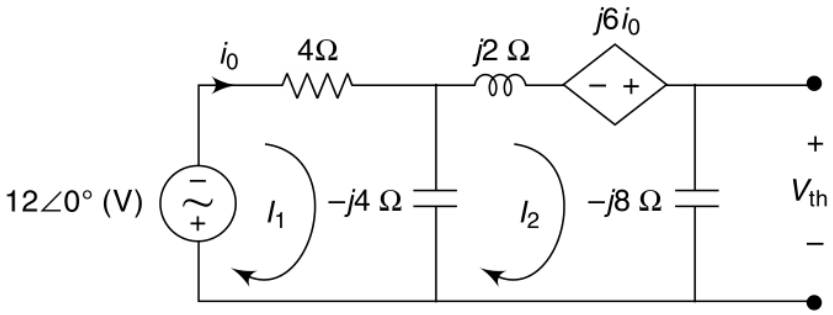
Find $V_0$ using Thevenin's theorem.




ω = 1 rad/s. The source's + is at the bottom; $i_0$ flows right through the 4 Ω.

**TRY IT FIRST**

$\omega = 1$: work out the three reactances first. Transform $3i_0 \parallel j2$ into a voltage source, then write two mesh equations for each of $V_{th}$ and $I_N$.




2 Ω removed; $3i_0 \parallel j2\ \Omega \rightarrow j6i_0$ in series with j2 Ω (+ on the right)

**Reactances ($\omega = 1$):** 2 H → j2 Ω, $\frac{1}{4}$ F → −j4 Ω, $\frac{1}{8}$ F → −j8 Ω.

**$V_{th}$** (2 Ω removed; clockwise mesh currents, $i_0 = I_1$):

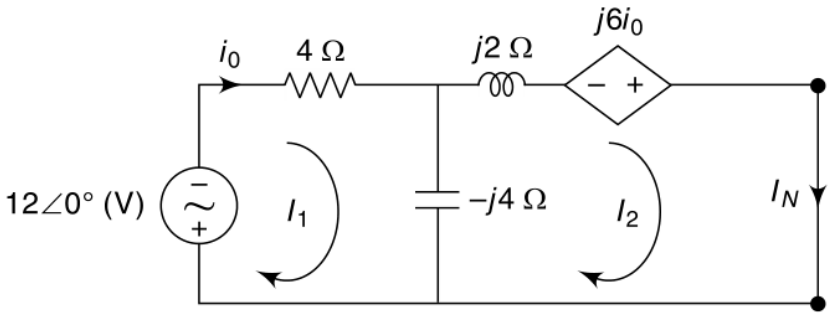
$$\text{Mesh 1:} \qquad (4 - j4)I_1 + j4I_2 = -12$$

$$\text{Mesh 2:} \qquad -j4(I_2 - I_1) + j2I_2 - j6I_1 - j8I_2 = 0 \quad \Rightarrow \quad -j2I_1 - j10I_2 = 0$$

$$\Rightarrow \quad I_2 = -0.2I_1$$

$$(4 - j4.8)I_1 = -12 \quad \Rightarrow \quad I_1 = -1.230 - j1.475\ \text{A}, \qquad I_2 = 0.246 + j0.295\ \text{A}$$

$$V_{th} = I_2(-j8) = 2.361 - j1.967 = 3.073\angle -39.81°\ \text{V}$$




Terminals shorted (the −j8 is shorted out)

**$I_N$** (terminals shorted):

$$\text{Mesh 2:} \qquad -j4(I_2 - I_1) + j2I_2 - j6I_1 = 0 \quad \Rightarrow \quad I_2 = -I_1; \qquad (4 - j8)I_1 = -12$$

$$I_N = I_2 = \frac{12}{4 - j8} = 0.6 + j1.2 = 1.342\angle 63.43^\circ \text{ A}$$

$$Z_{th} = \frac{V_{th}}{I_N} = 2.290\angle -103.24^\circ = -0.525 - j2.230\ \Omega$$

$$V_0 = V_{th} \times \frac{2}{Z_{th} + 2} = \frac{3.073\angle -39.81^\circ \times 2}{2.674\angle -56.50^\circ} = 2.299\angle 16.70^\circ \text{ V}$$

**ANSWER**

**$V_0 = 2.30\angle 16.70^\circ$ V, i.e. $v_0(t) = 2.30\cos(t + 16.70^\circ)$ V**

**BOOK ERROR IN 5.22**

For $V_{th}$, the book writes mesh 2 as $-j2I_1 - j6I_2 = 0$: it keeps the shared $-j4$ in the $I_1$ term but drops it from $I_2$'s own impedance ($-j4 + j2 - j8 = -j10$, not $-j6$). That gives $V_{th} = 4.8\angle -36.87^\circ$ V, $Z_{th} = 3.58\angle -100.3^\circ\ \Omega$ and $V_0 = 1.27\angle 32^\circ$ V. Its $I_N = 1.342\angle 63.43^\circ$ A is right. The answer above was confirmed by a full node solve with the 2 Ω in place ($2.2988\angle 16.70^\circ$ V).

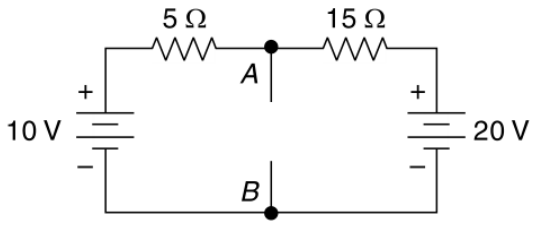
---

## A23 · Problem 5.23

Norton · DC **new idea**

**QUESTION**

Obtain the Norton's equivalent circuit with respect to the terminals AB for the network shown in the figure.

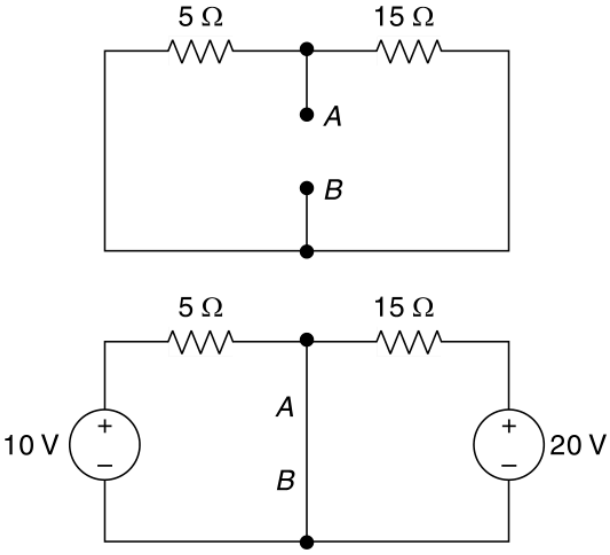



**TRY IT FIRST**

Short A to B. What current does each source push through the short?

**HINT: WHAT'S NEW**

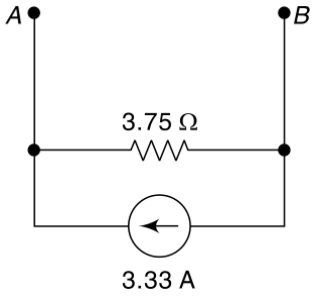
- **Norton's form:** $I_N = I_{sc}$ (current through the A–B short), $R_N = R_{th}$, drawn as a current source in parallel with $R_N$, arrow pointing towards A.
- With A–B shorted, each source sees only its own resistor, so $I_{sc}$ is a sum of two Ohm's-law currents.




Top: sources removed ($R_N$). Bottom: A–B shorted ($I_N$).

$$R_N = 5 \parallel 15 = \frac{75}{20} = 3.75\ \Omega$$

$$I_N = I_{sc} = \frac{10}{5} + \frac{20}{15} = 2 + 1.333 = 3.33\ \text{A}$$




Norton equivalent (arrow towards A)

**ANSWER**

**$I_N$ = 3.33 A (towards A) in parallel with $R_N$ = 3.75 Ω** (= book). Check: $V_{th} = I_N R_N$ = 12.5 V = node voltage at A ✓.

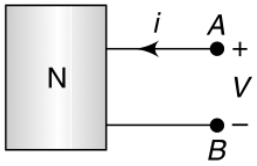
---

## A24 · Problem 5.24

Norton · from a v–i characteristic **new idea**

**QUESTION**

For the one-port shown in the figure, determine the Norton's equivalent at the terminals AB, if the v–i characteristic is given by 16v = 80 − 2i.




**TRY IT FIRST**

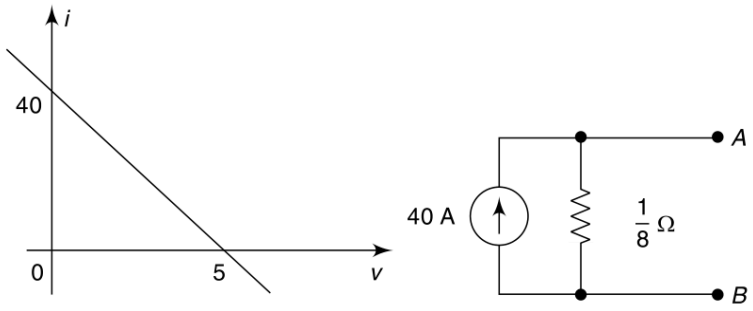
Which two points on the v–i line are $V_{oc}$ and $I_{sc}$?

**HINT: WHAT'S NEW**

- **The equivalent is read off the terminal line.** i = 0 gives $V_{oc}$; v = 0 gives $I_{sc}$; $R_N = V_{oc} / I_{sc}$ (the size of the line's slope, Δv/Δi).

$$16v = 80 - 2i \qquad \Rightarrow \qquad \frac{v}{5} + \frac{i}{40} = 1$$

$$i = 0: \quad V_{oc} = 5\ V \qquad v = 0: \quad I_{sc} = 40\ A \qquad R_N = \frac{5}{40} = \frac{1}{8}\ \Omega$$




The v–i line and the Norton equivalent

**ANSWER**

**$I_N$ = 40 A (towards A) in parallel with $R_N$ = 1/8 Ω** (= book)

**SIGN CONVENTION**

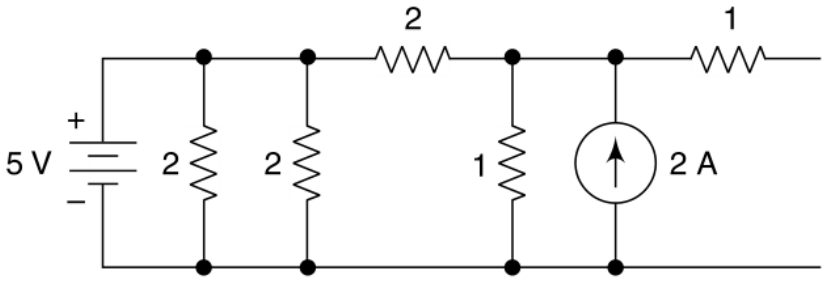
$v = 5 - \frac{i}{8}$ describes an ordinary source only if i is the current **delivered out of A** (v falls as the load draws more). The figure's arrow points into N; read literally, that would make $R_N = -1/8\ \Omega$ with the 40 A reversed. The intended answer is the book's.

## A25 · Problem 5.25

Thevenin + Norton · superposition inside **practice**

**QUESTION**

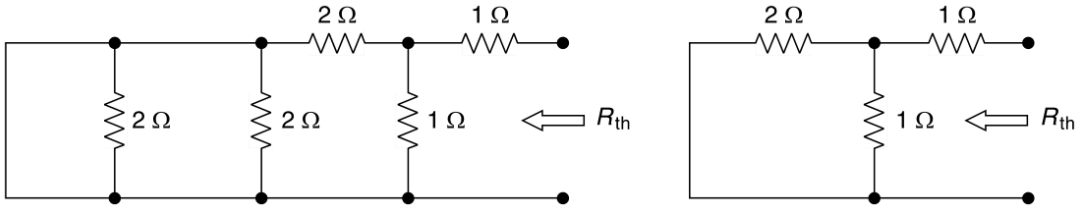
Find both Thevenin's and Norton's equivalent circuits for the network shown in the figure. All values are in ohm.




Terminals at the right-hand end of the 1 Ω series resistor and the bottom rail.

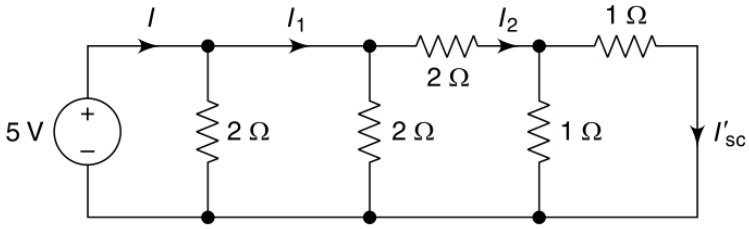
**TRY IT FIRST**

Which resistors does the shorted 5 V source remove from $R_{th}$? Then find $I_{sc}$ one source at a time.




Sources switched off: the two 2 Ω across the 5 V are shorted out

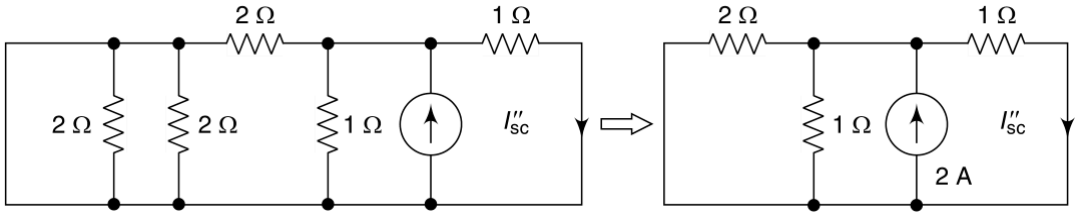
$$R_N = R_{th} = (2 \parallel 1) + 1 = \frac{2}{3} + 1 = \frac{5}{3}\ \Omega$$




5 V alone, terminals shorted

**$I_{sc}$, 5 V alone** (2 A opened): the middle node sees 2 Ω to the 5 V, and 1 Ω ∥ 1 Ω to ground:

$$\frac{V_m - 5}{2} + \frac{V_m}{1} + \frac{V_m}{1} = 0 \quad \Rightarrow \quad V_m = 1\ \text{V}, \qquad I_{sc}' = \frac{1}{1} = 1\ \text{A}$$

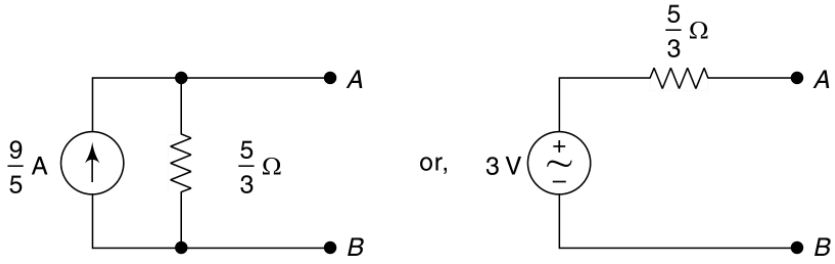



2 A alone, terminals shorted

**$I_{sc}$, 2 A alone** (5 V shorted): the 2 A divides between 2 Ω ‖ 1 Ω = 2/3 Ω and the 1 Ω to the short:

$$I_{sc}'' = 2 \times \frac{2/3}{2/3 + 1} = \frac{4}{5}\ \text{A}$$

$$I_N = 1 + \frac{4}{5} = \frac{9}{5}\ \text{A}, \qquad V_{th} = I_N R_{th} = \frac{9}{5} \times \frac{5}{3} = 3\ \text{V}$$




Norton and Thevenin forms

**ANSWER**

**Norton: 1.8 A ‖ 5/3 Ω. Thevenin: 3 V in series with 5/3 Ω** (= book)

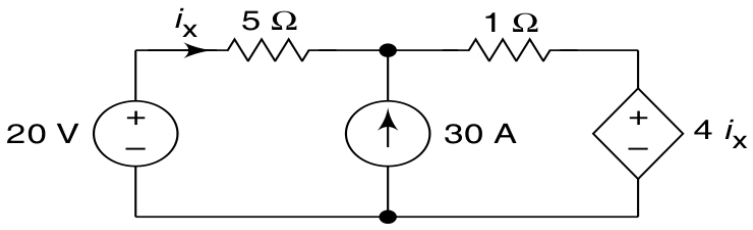
---

## A26 · Problem 5.26

Thevenin · dependent source **practice**

**QUESTION**

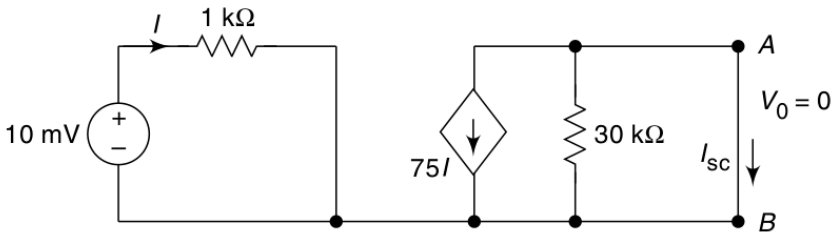
Replace the circuit in the figure with the Thevenin's equivalent circuit across A and B.




The question's figure, as printed: it marks no terminals A and B.

**BOOK ERROR IN 5.26: FIGURE AND SOLUTION BELONG TO DIFFERENT CIRCUITS**

The figure shows no A and B, so the question cannot be answered as printed. The book's solution solves a different circuit (a transistor model): 10 mV source, 1 kΩ carrying I, a controlled voltage $10^{-4}V_0$ in the input loop, and a controlled current source 75I in parallel with 30 kΩ across A–B. That solution is checked below. Ask your professor which circuit is meant before submitting.




The book's solution circuit, A–B shorted

**$V_{oc}$** (A–B open). The 75I pulls current down out of node A, and it can only return through the 30 kΩ, so $V_0 = 30k \times (-75I) = -2.25 \times 10^6 I$. Input loop:

$$10^3 I + 10^{-4}V_0 = 10 \times 10^{-3} \quad \Rightarrow \quad -4.444 \times 10^{-4}V_0 + 1 \times 10^{-4}V_0 = 10^{-2}$$
$$\Rightarrow \quad V_0 = -29.03 \text{ V}$$

**$I_{sc}$.** A–B shorted $\Rightarrow V_0 = 0$, so $I = \dfrac{10 \text{ mV}}{1 \text{ k}\Omega} = 10^{-5}$ A and $I_{sc} = -75I = -7.5 \times 10^{-4}$ A.

$$R_{th} = \frac{-29.03}{-7.5 \times 10^{-4}} = 38.71 \text{ k}\Omega$$

**ANSWER**

**(solution circuit) $V_{th} = -29.0$ V, $R_{th} = 38.7$ kΩ** (≈ book: it rounds $V_0$ to −29 V and prints 38.67 kΩ)

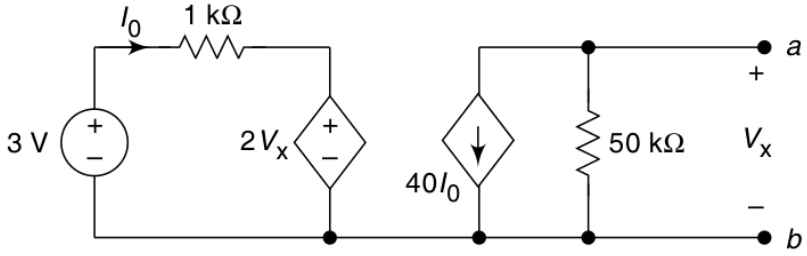
---

## A27 · Problem 5.27

Thevenin · dependent sources, negative $R_{th}$ **practice**

**QUESTION**

Find the Thevenin's equivalent between terminals a and b of the circuit shown in the figure.




The question figure gives 50 kΩ; the book's solution figure uses 50 Ω.

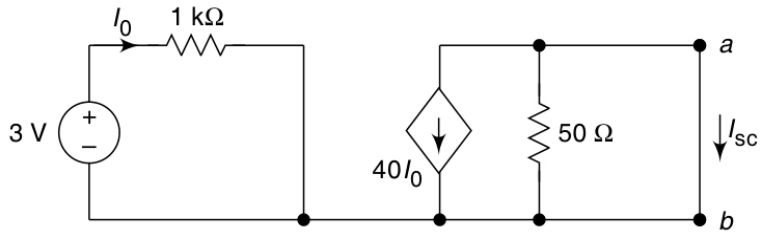
**TRY IT FIRST**

$V_x = V_{oc}$ feeds back into the left loop. For $I_{sc}$, $V_x = 0$ removes the $2V_x$ source.

**$V_{oc}$** ($R = 50$ kΩ, as in the question figure). Left loop and right node:

$$I_0 = \frac{3 - 2V_x}{1000}, \qquad V_{oc} = V_x = -40I_0 \times 50000 = -2000(3 - 2V_{oc})$$

$$3999V_{oc} = 6000 \qquad \Rightarrow \qquad V_{oc} = 1.5004 \text{ V}$$




a–b shorted

**$I_{sc}$** ($V_x = 0 \Rightarrow 2V_x = 0$): $I_0 = \frac{3}{1000} = 3$ mA. The 50 kΩ is shorted, and the dependent source pulls $40I_0$ from a down to b, so the short carries it from b to a:

$$I_{sc}\ (a \to b) = -40I_0 = -0.12 \text{ A}, \qquad R_{th} = \frac{1.5004}{-0.12} = -12.50\ \Omega$$

**ANSWER**

**$V_{th} = 1.500$ V, $R_{th} = -12.50\ \Omega$** (figure value 50 kΩ)

**BOOK ERRORS IN 5.27**

1. The solution uses **50 Ω** ($V_x = -40I_0 \times 50 = -2000I_0$), not the figure's 50 kΩ. With 50 Ω: $V_{oc} = -2(3 - 2V_{oc}) \Rightarrow V_{oc} = 2$ V.
2. It then writes $R_{th} = 2/0.12 = 16.67\ \Omega$, dropping the sign of $I_{sc} = -0.12$ A. With 50 Ω the correct value is **$R_{th} = -16.67\ \Omega$**.

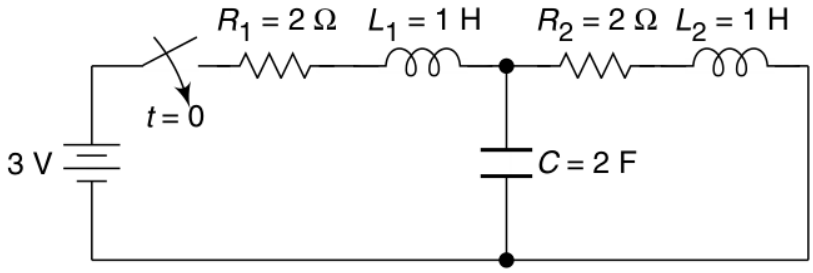
Test-source check (3 V shorted, 1 V applied at a): $I_0 = -\frac{2}{1000}$ A, current drawn $= \frac{1}{R} - 0.08$, giving $R_{th} = -12.50\ \Omega$ (50 kΩ) or $-16.67\ \Omega$ (50 Ω) ✓.

## A28 · Problem 5.28

Norton · s-domain transient **new idea**

**QUESTION**

In the network shown in the figure, the switch is closed at time t = 0. Assuming all the initial currents and voltages as zero, find the current through the inductor $L_2$ by the use of Norton's theorem.

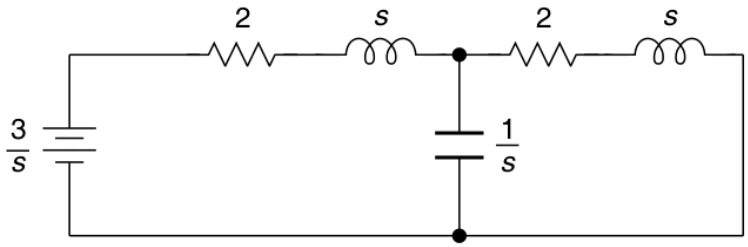



**TRY IT FIRST**

Draw the s-domain circuit (zero initial conditions). Make ($R_2$ + $sL_2$) the load and find the Norton equivalent of the rest.

**HINT: WHAT'S NEW**

- **Transient = theorem in the s-domain.** A DC source switched on at t = 0 becomes $\frac{V}{s}$; R → R, L → sL, C → $\frac{1}{sC}$. Find $I_L(s)$ exactly as in AC, then invert with partial fractions.
- With A–B shorted the capacitor is shorted out, so $I_{sc}$ is one Ohm's-law step.




The book's s-domain network: note it draws the capacitor as 1/s

**s-domain:** source $\frac{3}{s}$; $R_1 + sL_1 = s + 2$; C = 2 F → $\frac{1}{2s}$; load $R_2 + sL_2 = s + 2$.

$$Z_N = (s+2) \parallel \frac{1}{2s} = \frac{s+2}{2s^2+4s+1}, \qquad I_N = I_{sc} = \frac{3/s}{s+2} = \frac{3}{s(s+2)}$$

$$I_L(s) = I_N \times \frac{Z_N}{Z_N + (s+2)} = \frac{3}{s(s+2)} \times \frac{1}{1 + (2s^2+4s+1)} = \frac{3}{2s(s+2)(s+1)^2}$$

Partial fractions: $I_L(s) = \frac{3/4}{s} - \frac{3/4}{s+2} + \frac{0}{s+1} - \frac{3/2}{(s+1)^2}$

(Coefficients: at $s = 0$, $\frac{3}{2\cdot2\cdot1} = \frac{3}{4}$; at $s = -2$, $\frac{3}{2(-2)(1)} = -\frac{3}{4}$; on $(s+1)^2$, $\frac{3}{2(-1)(1)} = -\frac{3}{2}$; on $1/(s+1)$, 0, because the $s^3$ terms must cancel.)

**ANSWER**

$\mathbf{i_{L2}(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}te^{-t}\ A}$ $(t \geq 0)$

**BOOK ERROR IN 5.28**

The figure gives C = 2 F, but the book's s-domain network uses $\frac{1}{s}$ (that is C = 1 F) and prints $i(t) = \frac{3}{4} - \frac{3}{4}e^{-2t} - \frac{3}{2}e^{-t}$ sin t A. That answer is right for a 1 F capacitor only. A direct simulation of the circuit equations with C = 2 F matches the answer above (e.g. i(1 s) = 0.0967 A; the book's formula gives 0.184 A). Checks: $i(0) = 0$ and $i(\infty) = \frac{3}{2+2} = 0.75$ A.

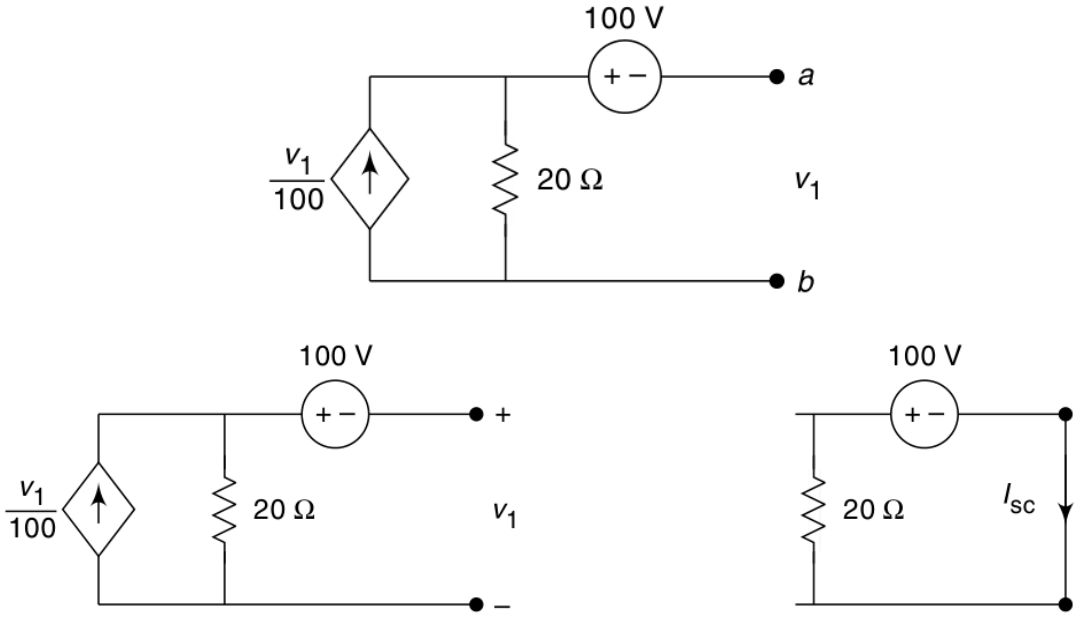
---

## A29 · Problem 5.29

Thevenin · dependent source **practice**

**QUESTION**

The following circuit has a dependent current source and an independent voltage source. Find the Thevenin equivalent network of the circuit across the terminals a and b.




Left: a–b open ($v_1 = v_{oc}$). Right: a–b shorted ($v_1 = 0$, so the dependent source is zero).

**$V_{oc}$:** the top node sits at $v_1 + 100$. KCL there:

$$\frac{v_1}{100} = \frac{v_1 + 100}{20} \quad \Rightarrow \quad v_1 = 5v_1 + 500 \quad \Rightarrow \quad V_{oc} = -125\ \text{V}$$

**$I_{sc}$:** $v_1 = 0$, so the dependent source is off; the top node is at 100 V and the 20 Ω draws 5 A from it, which must come from the short: $I_{sc}$ (a → b) = −5 A.

$$R_{th} = \frac{-125}{-5} = 25\ \Omega$$

**ANSWER**

**$V_{th}$ = −125 V (i.e. 125 V with + at b), $R_{th}$ = 25 Ω** (= book)

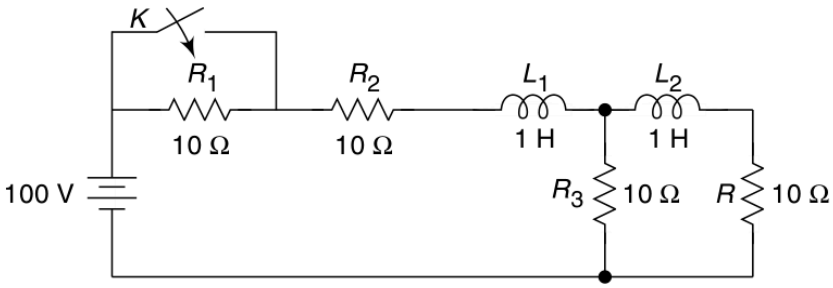
---

## A30 · Problem 5.30

Thevenin · transient with initial conditions **new idea**

**QUESTION**

In the network of the figure, the switch K is closed at time t = 0, a steady state having previously existed. Obtain the current in the resistor R using Thevenin's theorem.



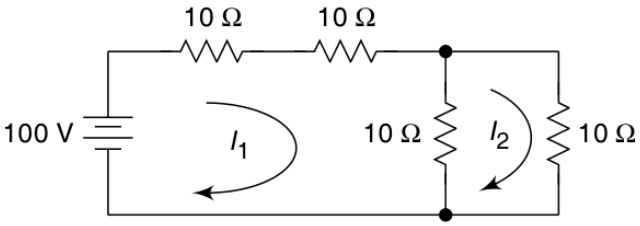

Closing K shorts $R_1$.

**TRY IT FIRST**

Find both inductor currents just before t = 0 (inductors are shorts in DC steady state). Then build the s-domain circuit with those currents as sources.

**HINT: WHAT'S NEW**

- **Non-zero initial current:** an inductor carrying i(0) becomes sL **in series with a voltage source L·i(0)**, polarity aiding the current's direction.



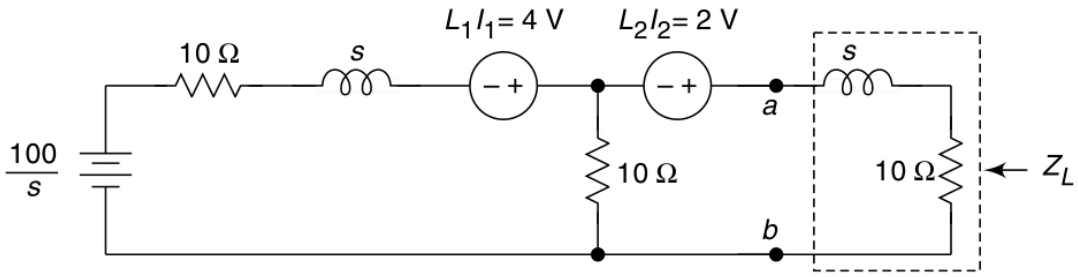

*(a) Circuit at $t = 0-$*

t = 0−: inductors as shorts

**t = 0−** (K open, $R_1$ in):

$$30I_1 - 10I_2 = 100, \qquad -10I_1 + 20I_2 = 0$$

$$i_{L1}(0) = I_1 = 4\ \text{A}, \qquad i_{L2}(0) = I_2 = 2\ \text{A}$$




*(b) Transform network for $t > 0$*

s-domain network for t > 0 ($R_1$ shorted); the load $Z_L = s + 10$ is in the dashed box

**Thevenin at a–b** (load = $sL_2 + R = s + 10$; the 2 V of $L_2$ stays on the source side):

$$Z_{th} = (s + 10) \parallel 10 = \frac{10(s + 10)}{s + 20}$$

$$I(s) = \frac{100/s + 4}{s + 20} = \frac{4s + 100}{s(s + 20)}, \qquad V_{th} = 10I(s) + 2 = \frac{2s^2 + 80s + 1000}{s(s + 20)}$$

$$I_R(s) = \frac{V_{th}}{Z_{th} + s + 10} = \frac{2s^2 + 80s + 1000}{s(s + 10)(s + 30)} = \frac{10/3}{s} - \frac{2}{s + 10} + \frac{2/3}{s + 30}$$

**ANSWER**

$$\mathbf{i_R(t) = \frac{10}{3} - 2e^{-10t} + \frac{2}{3}e^{-30t}\ A = 3.33 - 2e^{-10t} + 0.67e^{-30t}\ A}\ (= \text{book})$$

**CHECK**

$i(0) = 10/3 - 2 + 2/3 = 2\text{ A} = i_{L2}(0)$ ✓. $i(\infty) = \dfrac{100}{10+5} \times \dfrac{1}{2} = 3.33\text{ A}$ ✓. A direct simulation of the circuit equations agrees at every time tested.

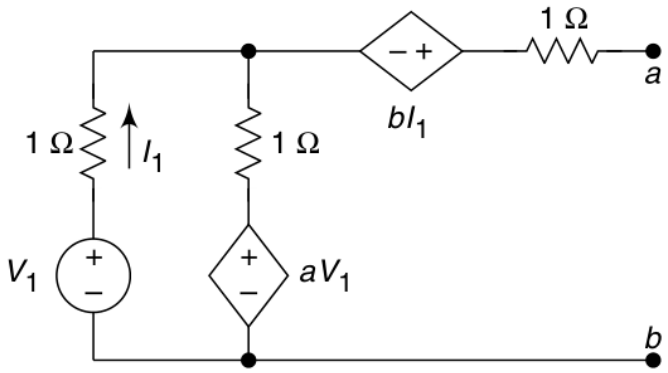
---

## A31 · Problem 5.31

Thevenin · symbolic, two dependent sources **new idea**

**QUESTION**

For the network shown in the figure, show that the Thevenin equivalent at the terminals a-b is represented by $V_{Th} = \dfrac{V_1}{2}(1 + a + b - ab)$ and $Z_{Th} = \dfrac{3 - b}{2}$.




**TRY IT FIRST**

Keep a, b and $V_1$ as letters. Open circuit: no current in the right 1 Ω. Short circuit: two mesh equations.

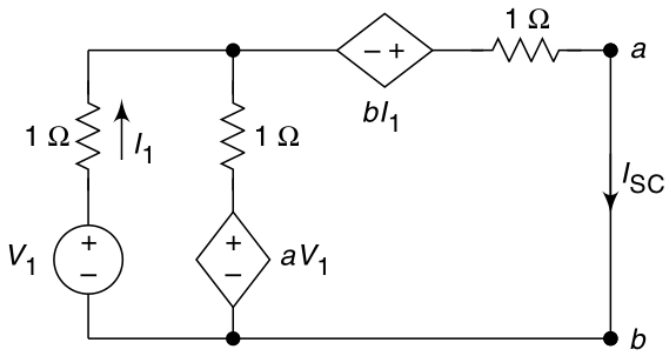
**HINT: WHAT'S NEW**

- **Symbolic answer:** the method is unchanged; carry the letters and simplify at the end. Substituting numbers (e.g. a = b = 0) checks the algebra.

**$V_{th}$** (no current in the right 1 Ω). Left mesh:

$$2I_1 + aV_1 = V_1 \qquad \Rightarrow \qquad I_1 = \frac{V_1}{2}(1 - a)$$

$$V_{th} = aV_1 + 1 \cdot I_1 + bI_1 = aV_1 + (1 + b)\frac{V_1}{2}(1 - a) = \frac{V_1}{2}(1 + a + b - ab)$$




a–b shorted

**$I_{sc}$.** Two meshes:

$$2I_1 - I_{SC} = V_1(1 - a) \quad \text{...(i)} \qquad -(1 + b)I_1 + 2I_{SC} = aV_1 \quad \text{...(ii)}$$

$$I_{SC} = \frac{2aV_1 + (1 + b)V_1(1 - a)}{4 - (1 + b)} = \frac{V_1(1 + a + b - ab)}{3 - b}$$

$$Z_{th} = \frac{V_{th}}{I_{SC}} = \frac{3 - b}{2} \qquad \text{(proved)}$$

**CHECK**

Test source ($V_1 = 0$): the top node sits on $1 \parallel 1 = 0.5\ \Omega$, $I_1 = -0.5I_t$, so $V = I_t + 0.5I_t - 0.5bI_t = \frac{3 - b}{2} I_t$ ✓.

Also confirmed numerically for two sets of (a, b, $V_1$).

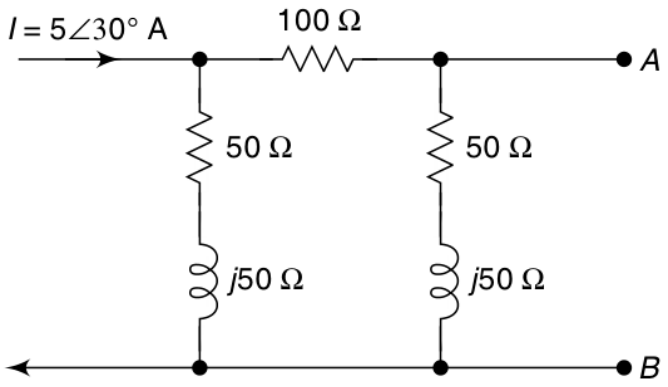
---

## A32 · Problem 5.32

Thevenin · AC, current source **practice**

**QUESTION**

Find the Thevenin equivalent circuit for the network shown in the figure at terminals A-B.




The 5∠30° A is fed in at the left node and returns on the bottom rail.

$$I_{right} = 5\angle 30° \times \frac{50 + j50}{(100 + 50 + j50) + (50 + j50)} = 5\angle 30° \times \frac{1 + j}{4 + j2} = 1.581\angle 48.43° \text{ A}$$

$$V_{th} = I_{right}(50 + j50) = 111.8\angle 93.43° \text{ V}$$

$$Z_{th} = (150 + j50) \parallel (50 + j50) = \frac{5000 + j10000}{200 + j100} = 40 + j30\ \Omega$$

**ANSWER**

**$V_{th} = 111.8\angle 93.43°$ V, $Z_{th} = 50\angle 36.87° = 40 + j30\ \Omega$** (= book)

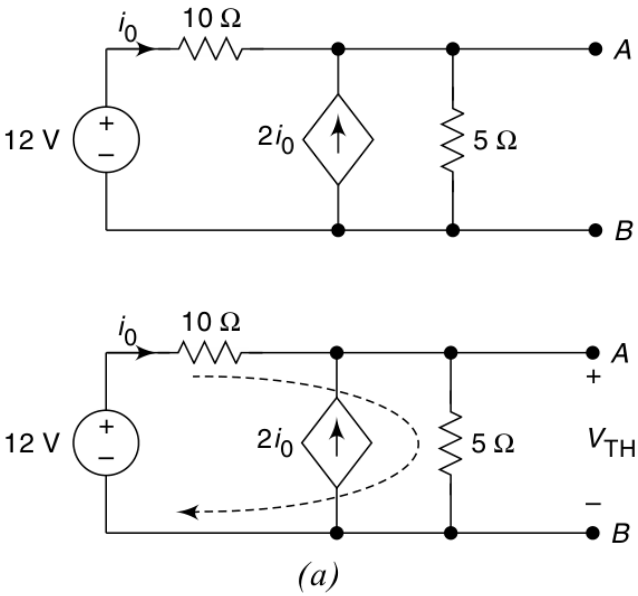
---

## A33 · Problem 5.33

Thevenin · dependent source **practice**

**QUESTION**

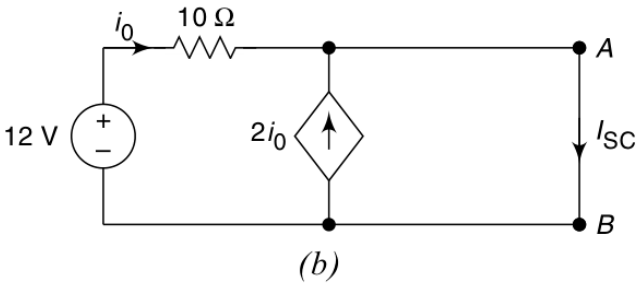
Find Thevenin's equivalent circuit across the terminals A and B for the network shown in the figure.




A–B open

KVL: $10i_0 + V_{th} = 12$ KCL at A: $i_0 + 2i_0 = \frac{V_{th}}{5}$ ⇒ $V_{th} = 15i_0$

$25i_0 = 12$ ⇒ $i_0 = 0.48$ A, $V_{th} = 7.2$ V




A–B shorted (the 5 Ω is shorted out)

$$i_0 = \frac{12}{10} = 1.2\ \text{A}, \qquad I_{sc} = i_0 + 2i_0 = 3.6\ \text{A}, \qquad R_{th} = \frac{7.2}{3.6} = 2\ \Omega$$

**ANSWER**

**$V_{th}$ = 7.2 V, $R_{th}$ = 2 Ω** (= book)

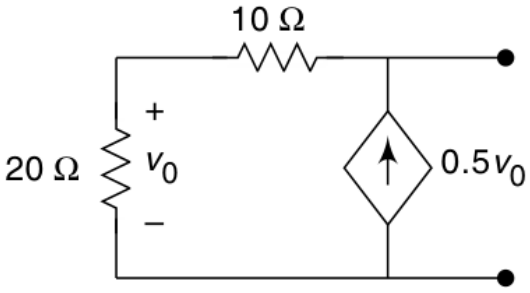
---

## A34 · Problem 5.34

Thevenin · no independent source **new idea**

**QUESTION**

Find Thevenin's equivalent circuit for the network shown in the figure.

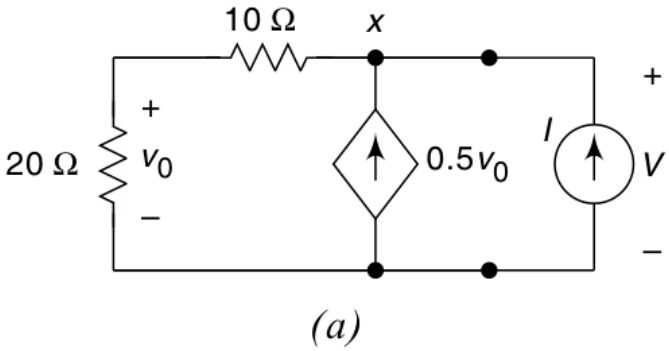



**TRY IT FIRST**

There is nothing to drive the circuit. What is $V_{th}$? For $Z_{th}$, connect a 1 A source and find the voltage it produces.

**HINT: WHAT'S NEW**

- **Only dependent sources:** $V_{th}$ = 0 (no independent source, no response), and $I_{sc}$ = 0 too, so $V_{oc}/I_{sc}$ fails. Use a **test source**: $Z_{th}$ = V / I.




*(a)*

Test current I applied; V measured

KCL at x: $\dfrac{V - v_0}{10} - 0.5v_0 - I = 0 \quad \Rightarrow \quad V = 10I + 6v_0$

Current down the 20 Ω = $\dfrac{V - v_0}{10} = 0.5v_0 + I$

$v_0 = 20(0.5v_0 + I) \quad \Rightarrow \quad v_0 = -\dfrac{20}{9}I$

$V = 10I - \dfrac{120}{9}I = -\dfrac{10}{3}I \quad \Rightarrow \quad Z_{th} = -3.33\ \Omega$

**ANSWER**

**$V_{th} = 0$, $Z_{th} = -10/3 = -3.33\ \Omega$** (= book; its final sketch omits the minus sign)

## C · Maximum power transfer theorem

Thevenin / Norton (done) → Load sees only $V_{th}$ and $Z_{th}$ → **Maximum power: choose the load to match $Z_{th}$**

**Where this sits.** Thevenin's theorem reduced any network to $V_{th}$ in series with $Z_{th}$. The load current is then $V_{th} / (Z_{th} + Z_L)$, so the only question left is which load draws the most power from that one source.

**Three terms first.**

- **Load:** the element connected at the terminals, $R_L$ (DC) or $Z_L = R_L + jX_L$ (AC).
- **Average power in the load:** $P = |I|^2 R_L$. Only the resistive part of the load absorbs power.
- **Complex conjugate:** for $Z_{th} = R_{th} + jX_{th}$, the conjugate is $Z_{th}^* = R_{th} - jX_{th}$ (same resistance, opposite reactance).

**Statement.**

| Load allowed to vary | Maximum power when | $P_{max}$ |
|---|---|---|
| $R_L$ only, DC | $R_L = R_{th}$ | $\frac{V_{th}^2}{4R_{th}}$ |
| $Z_L = R_L + jX_L$, both free (AC) | $Z_L = Z_{th}^*$ (reactances cancel) | $\frac{\lvert V_{th}\rvert^2}{4R_{th}}$ |
| Pure resistance $R_L$, AC source | $R_L = \lvert Z_{th}\rvert$ | $\lvert I\rvert^2 R_L$ (calculate) |

The recipe is always: **find $V_{th}$ and $Z_{th}$, then match.** At maximum power the load gets exactly half the power, so efficiency is only 50%.

**Phasor size convention.** The book takes phasor magnitudes as rms values, so $P = |V_{th}|^2 / (4R_{th})$. If a source is given as a peak value, the power is half of that.

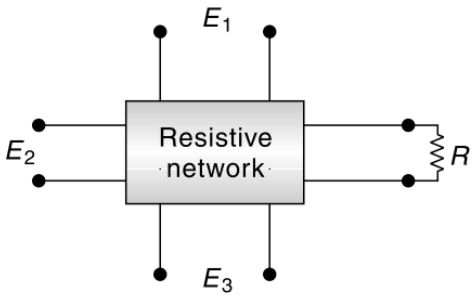
---

## A35 · Problem 5.35

Power and superposition **new idea**

**QUESTION**

In the network shown, the power dissipated in R when $E_1$, $E_2$ or $E_3$ acting alone is (a) 20 W, 80 W and 5 W respectively; (b) 30 W, 270 W and 120 W respectively. Calculate the maximum power that R can dissipate due to the simultaneous action of all the sources. Calculate both for (a) and (b). What will be the minimum power dissipated in R when all the sources are acting simultaneously?




**TRY IT FIRST**

Power does not add. What does add? Turn each power back into a current, and remember you don't know each current's sign.

**HINT: WHAT'S NEW**

- **Superposition adds currents, not powers.** Each source alone gives $|i_k| = \sqrt{(P_k/R)}$; the directions are unknown, so the total is $\pm i_1 \pm i_2 \pm i_3$.
- $P = i^2R = (\pm\sqrt{(P_1)} \pm \sqrt{(P_2)} \pm \sqrt{(P_3)})^2$. Largest: all signs the same. Smallest: the sign choice that makes the sum closest to zero.

$$i = \pm\sqrt{\left(\frac{P_1}{R}\right)} \pm \sqrt{\left(\frac{P_2}{R}\right)} \pm \sqrt{\left(\frac{P_3}{R}\right)} \quad \Rightarrow \quad P = i^2R = (\pm\sqrt{(P_1)} \pm \sqrt{(P_2)} \pm \sqrt{(P_3)})^2$$

**(a)** $\sqrt{(20)} = 2\sqrt{(5)}$, $\sqrt{(80)} = 4\sqrt{(5)}$, $\sqrt{(5)}$:

$$P_{max} = (2 + 4 + 1)^2 \times 5 = 49 \times 5 = 245 \text{ W}, \qquad P_{min} = (4 - 2 - 1)^2 \times 5 = 5 \text{ W}$$

**(b)** $\sqrt{(30)}$, $\sqrt{(270)} = 3\sqrt{(30)}$, $\sqrt{(120)} = 2\sqrt{(30)}$:

$$P_{max} = (1 + 3 + 2)^2 \times 30 = 36 \times 30 = 1080 \text{ W}, \qquad P_{min} = (3 - 1 - 2)^2 \times 30 = 0 \text{ W}$$

**ANSWER**

**(a) $P_{max}$ = 245 W, $P_{min}$ = 5 W. (b) $P_{max}$ = 1080 W, $P_{min}$ = 0 W** (= book)

**SIDE NOTE**

Adding the powers would give 105 W and 420 W, which is wrong in both cases. The book's working line for (b) $P_{max}$ misprints the bracket as $[4\sqrt{(5)} - 3\sqrt{(5)}]$; its answer 1080 W is right.

## A36 · Problem 5.36

Maximum power · DC **teach**

**QUESTION**

Find the value of R in the circuit of the figure such that maximum power transfer takes place. What is the amount of this power?

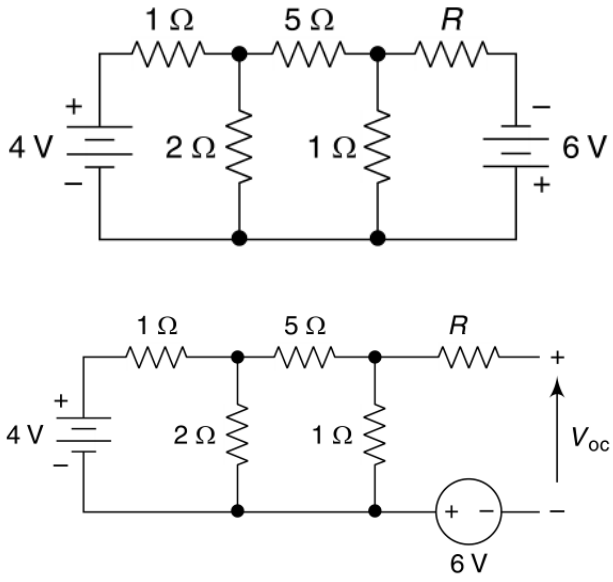
**TRY IT FIRST**

For each circuit: remove R, find $V_{th}$ across its terminals and $R_{th}$ looking back in. Then $R = R_{th}$ and $P_{max} = V_{th}^2/(4R_{th})$. In (b), look for an element that changes nothing. In (c), R is not connected to ground.

**HINT: WHAT'S NEW**

- **Maximum power transfer (DC):** $R = R_{th}$, $P_{max} = \dfrac{V_{th}^2}{4R_{th}}$. All the work is in finding $V_{th}$ and $R_{th}$.
- **(b)** the 2 Ω directly across the 5 V source changes no other current (5.1's shortcut); then source transformations reduce the rest.
- **(c)** R sits between two non-ground nodes, so $V_{th} = V_P - V_Q$.

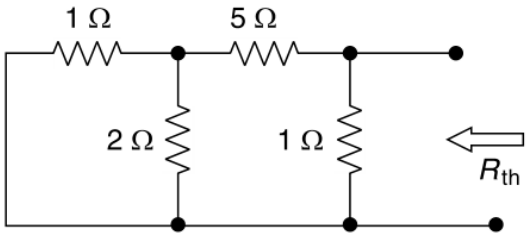
**(a)**




R removed; the 6 V source (+ at the bottom) is in the path to the terminal

**$V_{th}$.** Mesh currents with R removed: $3i_1 - 2i_2 = 4$, $-2i_1 + 8i_2 = 0 \Rightarrow i_1 = 4i_2$, so $10i_2 = 4$:

$$i_2 = \frac{2}{5}\ \text{A}, \qquad V_{th} = 1 \times i_2 + 6 = \frac{32}{5} = 6.4\ \text{V}$$




Sources shorted, looking back from R

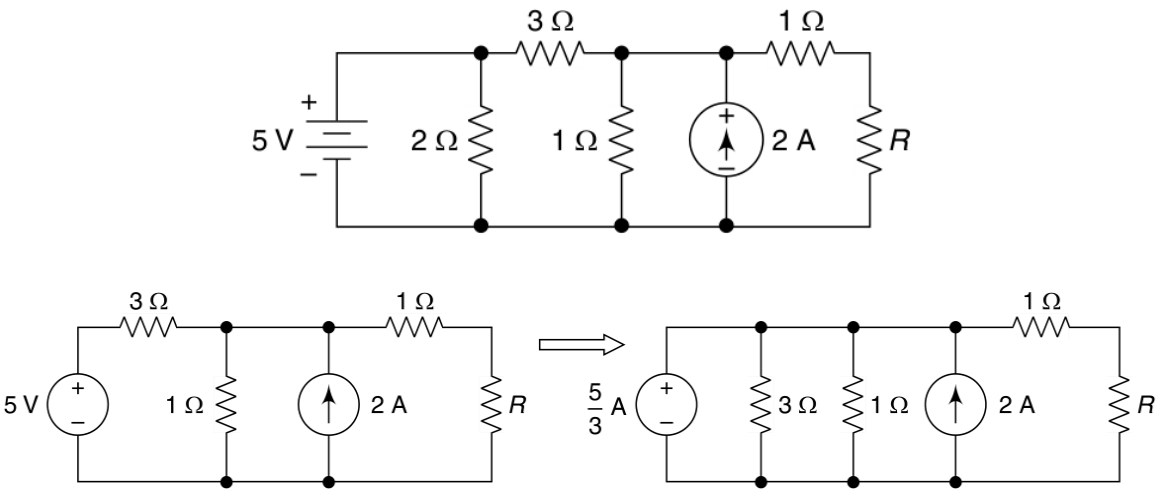
$$R_{th} = [(1 \parallel 2) + 5] \parallel 1 = \frac{17}{3} \parallel 1 = \frac{17}{20} = 0.85\ \Omega$$

$$P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{6.4^2}{4 \times 0.85} = \frac{40.96}{3.4} = 12.05\ \text{W}$$

**ANSWER**

**(a) R = 0.85 Ω, $P_{max}$ = 12.05 W** (= book, which rounds to 12 W)

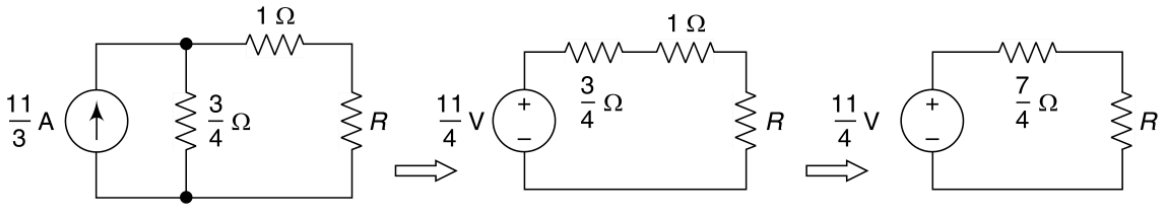
**(b)**




The 2 Ω across the 5 V is dropped; 5 V with 3 Ω → $\frac{5}{3}$ A ∥ 3 Ω

Combine the two current sources and the two resistors, then transform back:

$$\frac{5}{3} + 2 = \frac{11}{3}\ \text{A}, \qquad 3 \parallel 1 = \frac{3}{4}\ \Omega \qquad \Rightarrow \qquad \frac{11}{3} \times \frac{3}{4} = \frac{11}{4}\ \text{V in series with } \frac{3}{4}\ \Omega$$



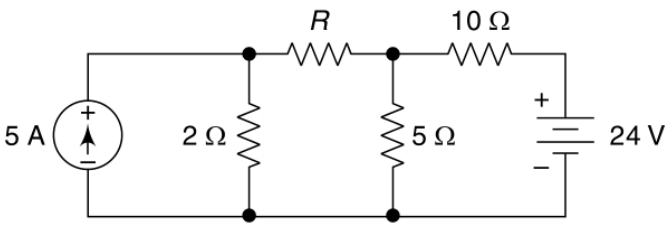

Successive source transformations

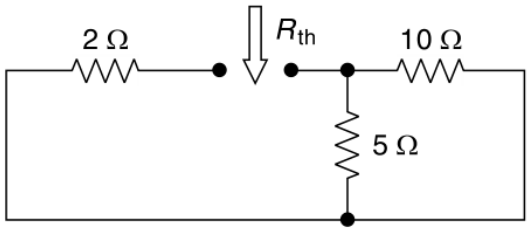
$$V_{th} = \frac{11}{4}\ \text{V}, \qquad R_{th} = \frac{3}{4} + 1 = \frac{7}{4}\ \Omega, \qquad P_{max} = \frac{(11/4)^2}{4 \times 7/4} = \frac{121}{112} = 1.08\ \text{W}$$

**ANSWER**

**(b) R = 7/4 = 1.75 Ω, $P_{max}$ = 1.08 W** (= book)

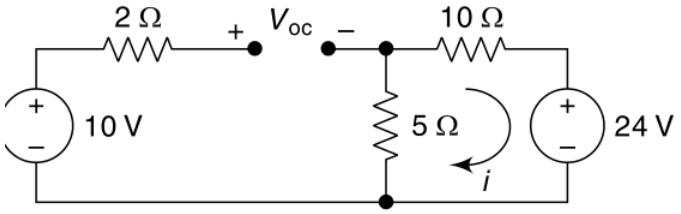
**(c)**




$R_{th}$: 5 A opened, 24 V shorted

$$R_{th} = 2 + (10 \parallel 5) = 2 + \frac{10}{3} = 5.33\ \Omega$$




$V_{oc}$: 5 A ‖ 2 Ω → 10 V in series with 2 Ω

**$V_{th}$.** With R removed, no current flows in the 2 Ω, so $V_P$ = 10 V. The right loop gives $V_Q = 24 \times \frac{5}{15} = 8$ V (the book writes this as i = −24/15 = −1.6 A, $V_{oc}$ = 5i + 10).

$$V_{th} = V_P - V_Q = 10 - 8 = 2\ \text{V}, \qquad P_{max} = \frac{2^2}{4 \times 16/3} = \frac{3}{16} = 0.1875\ \text{W}$$

**ANSWER**

**(c) R = 5.33 Ω, $P_{max}$ = 0.1875 W** (= book, 0.188 W)

**CHECK**

For each part, the full circuit was solved with R swept from 0 to 50 Ω; the power peaks at the R and $P_{max}$ above.

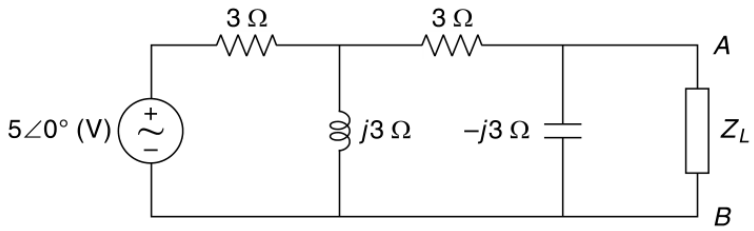
---

## A37 · Problem 5.37

Maximum power · AC conjugate match **new idea**

**QUESTION**

In the network shown, find the value of $Z_L$ to which the maximum power can be delivered. Hence, find the value of the maximum power.




**TRY IT FIRST**

Find $V_{th}$ and $Z_{th}$ at A–B as in any AC Thevenin problem. Then choose $Z_L$ so that the total impedance in the loop is purely resistive.

**HINT: WHAT'S NEW**

- **AC load free in R and X:** $Z_L = Z_{th}^*$. The reactances cancel ($X_L = -X_{th}$), leaving the DC case $R_L = R_{th}$, so $P_{max} = \dfrac{|V_{th}|^2}{4R_{th}}$.

**$V_{th}$.** Impedance seen by the source: $3 + j3 \parallel (3 - j3) = 3 + (3 + j3) = 6 + j3\ \Omega$.

$$I = \frac{5}{6 + j3}, \qquad I_{right} = I \times \frac{j3}{j3 + 3 - j3} = jI$$

$$V_{th} = I_{right}(-j3) = 3I = \frac{15}{6 + j3} = 2 - j1 = 2.236\angle -26.57°\ \text{V}$$

**$Z_{th}$** (source shorted):

$$3 \parallel j3 = 1.5 + j1.5, \qquad (1.5 + j1.5 + 3) \parallel (-j3) = \frac{(4.5 + j1.5)(-j3)}{4.5 - j1.5} = 1.8 - j2.4\ \Omega$$

$$Z_L = Z_{th}^{*} = 1.8 + j2.4\ \Omega, \qquad I = \frac{V_{th}}{2 \times 1.8} = 0.621\angle -26.57^\circ\ \text{A}$$

$$P_{max} = \frac{2.236^2}{4 \times 1.8} = 0.694\ \text{W}$$

**ANSWER**

**$Z_L = 1.8 + j2.4\ \Omega$, $P_{max} = 0.694$ W** (= book)

---

## A38 · Problem 5.38

Maximum power · AC load impedance **practice**

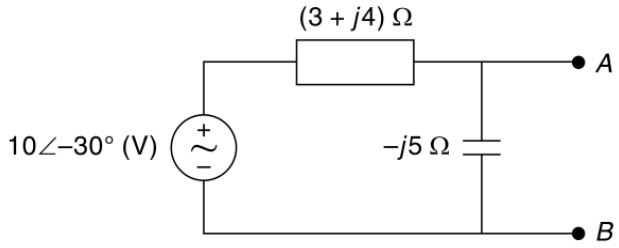
**QUESTION**

A loudspeaker is connected across terminals A and B of the network. What should be its impedance to obtain maximum power dissipation in it?

**TRY IT FIRST**

Only $Z_{th}$ is needed: switch the sources off and look back from A–B.

**(a)**

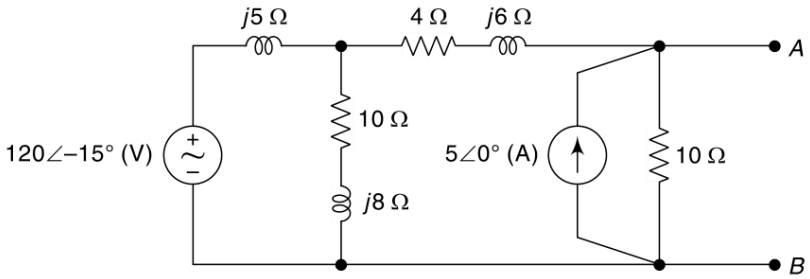



$$Z_{th} = \frac{(3 + j4)(-j5)}{3 + j4 - j5} = \frac{20 - j15}{3 - j1} = 7.5 - j2.5 = 7.91\angle -18.43^\circ\ \Omega$$

**ANSWER**

**(a) $Z_L = Z_{th}^{*} = 7.5 + j2.5\ \Omega$** (the book prints 7.5 + j2.4, a typo; its own $Z_{th}$ is 7.5 − j2.5)

**(b)**




Voltage source shorted and current source opened for $Z_{th}$.

$$(10 + j8) \parallel j5 = \frac{-40 + j50}{10 + j13} = 0.929 + j3.792\ \Omega$$

$$+ (4 + j6) = 4.929 + j9.792\ \Omega, \qquad \parallel 10\ \Omega \qquad \Rightarrow \qquad Z_{th} = 5.316 + j3.072 = 6.14\angle 30°\ \Omega$$

**ANSWER**

**(b) $Z_L = Z_{th}^{*} = 5.316 - j3.072 = 6.14\angle -30°\ \Omega$** (= book)

---

## A39 · Problem 5.39

Maximum power · resistive load on an AC source **new idea**

**QUESTION**

Two inductors each of 1 Ω reactance and negligible resistance are connected in series across a 2 V a.c. source. Find the value of resistance which should be connected across one of the inductors for maximum power dissipation. Also, find the maximum power.

**TRY IT FIRST**

Draw it: 2 V source, j1, then j1 ∥ R. Write the power in R as a function of R and maximise it, or find the Thevenin equivalent seen by R first.

**HINT: WHAT'S NEW**

- **Only a resistance is allowed**, so the reactance cannot be cancelled and $Z_L = Z_{th}^{*}$ is impossible. For a purely resistive load the best value is $R = |Z_{th}|$.

**Route 1: calculus** (as the book):

$$Z = \frac{jR}{R + j1} + j1 = \frac{-1 + j2R}{R + j1}, \qquad I_R = \frac{2(R + j1)}{-1 + j2R} \times \frac{j1}{R + j1} = \frac{j2}{-1 + j2R}$$

$$P = |I_R|^2 R = \frac{4R}{1 + 4R^2}, \qquad \frac{dP}{dR} = \frac{4(1 + 4R^2) - 4R(8R)}{(1 + 4R^2)^2} = 0 \qquad \Rightarrow \qquad 4 - 16R^2 = 0$$

$$\Rightarrow \qquad R = 0.5\ \Omega$$

$$P_{max} = \frac{4 \times 0.5}{1 + 4 \times 0.25} = 1\ \text{W}$$

**Route 2: Thevenin.** Seen by R: $V_{th} = 2 \times \frac{j1}{j1 + j1} = 1$ V, $Z_{th} = j1 \parallel j1 = j0.5\ \Omega$. Pure resistance $\Rightarrow R = |Z_{th}| = 0.5\ \Omega$, and $|I|^2 = \frac{1}{|0.5 + j0.5|^2} = 2$, so $P = 2 \times 0.5 = 1$ W ✓.

**ANSWER**

**$R = 0.5\ \Omega$, $P_{max} = 1$ W** (= book; its last line prints $4 \times 0.4$, a typo for $4 \times 0.5$)

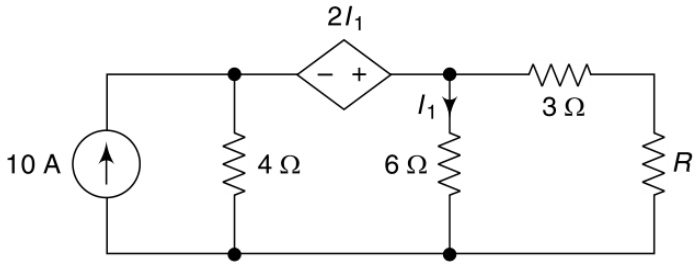
---

## A40 · Problem 5.40

Maximum power · dependent source **practice**

**QUESTION**

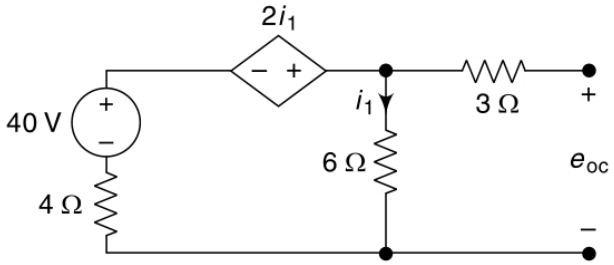
In the network shown, calculate the maximum power that may be dissipated in the external resistor R.




$2i_1$ is a current-controlled voltage source; $i_1$ flows down the 6 Ω.

**TRY IT FIRST**

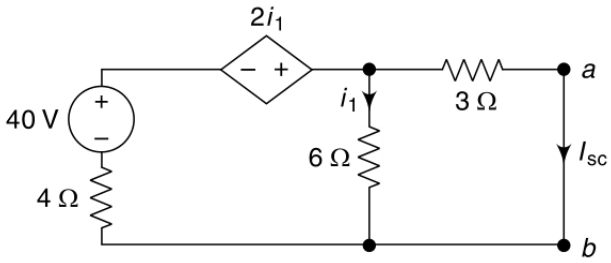
The dependent source rules out "switch off and combine", so find $V_{oc}$ and $I_{sc}$. Start by turning 10 A ∥ 4 Ω into a voltage source.




10 A ∥ 4 Ω → 40 V in series with 4 Ω; terminals open

**$V_{oc}$** (no current in the 3 Ω, so the loop current is $i_1$):

$$-40 + 4i_1 - 2i_1 + 6i_1 = 0 \quad \Rightarrow \quad i_1 = 5\ \text{A}, \quad V_{oc} = 6i_1 = 30\ \text{V}$$




Terminals shorted

**$I_{sc}$.** Let $V_M$ be the node above the 6 Ω: $i_1 = V_M/6$, $I_{sc} = V_M/3$, source current = $V_M/6 + V_M/3 = V_M/2$. Left loop:

$$40 - 4 \times \frac{V_M}{2} + 2 \times \frac{V_M}{6} - V_M = 0 \qquad \Rightarrow \qquad \frac{8}{3}V_M = 40$$

$$V_M = 15\ \text{V}, \qquad I_{sc} = \frac{15}{3} = 5\ \text{A}$$

$$R_{th} = \frac{30}{5} = 6\ \Omega, \qquad P_{max} = \frac{30^2}{4 \times 6} = 37.5\ \text{W}$$

**ANSWER**

**R = 6 Ω, $P_{max}$ = 37.5 W** (= book)

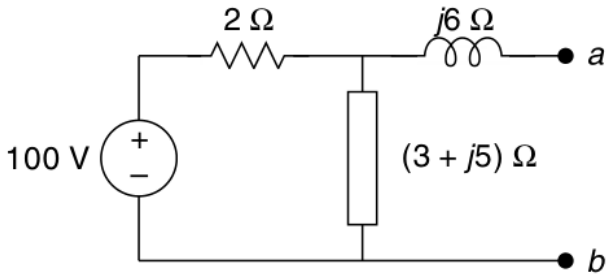
---

## A41 · Problem 5.41

Maximum power · AC **practice**

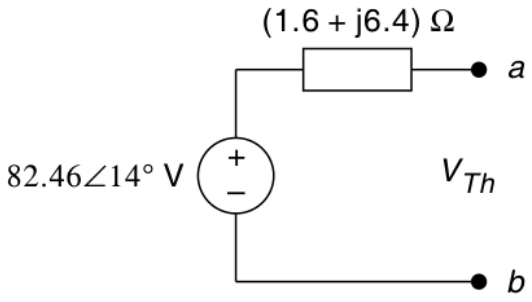
**QUESTION**

Find the Thevenin's equivalent between the points a and b for the circuit given in the figure. What should be the value of impedance connected between a and b for maximum power to be transferred from the sources? Obtain the amount of the maximum power.




$$I = \frac{100}{2 + 3 + j5} = 10 - j10\ \text{A}, \qquad V_{th} = I(3 + j5) = 80 + j20 = 82.46\angle 14.04°\ \text{V}$$

$$Z_{th} = j6 + \frac{2(3 + j5)}{5 + j5} = j6 + 1.6 + j0.4 = 1.6 + j6.4\ \Omega$$




Thevenin equivalent

$$Z_L = Z_{th}^{*} = 1.6 - j6.4\ \Omega, \qquad P_{max} = \frac{|V_{th}|^2}{4R_{th}} = \frac{6800}{4 \times 1.6} = 1062.5\ \text{W}$$

**ANSWER**

**$V_{th}$ = 82.46∠14.04° V, $Z_{th}$ = 1.6 + j6.4 Ω; $Z_L$ = 1.6 − j6.4 Ω, $P_{max}$ = 1062.5 W** (= book)

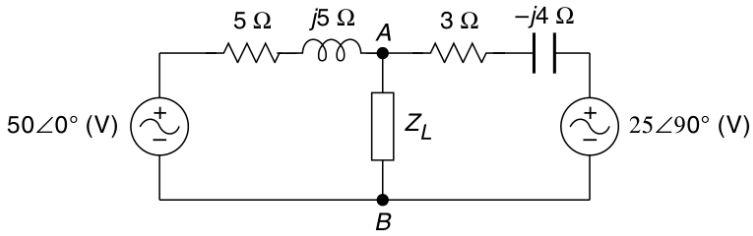
---

## A42 · Problem 5.42

Maximum power · AC, two sources **practice**

**QUESTION**

In the network in the figure, two voltage sources act on the load impedance connected to the terminal A and B. If the load is variable in both reactance and resistance, for what load $Z_L$ will receive maximum power? What is the value of maximum power?




**$V_{th}$** ($Z_L$ removed; one loop current from the 50 V towards the j25 V):

$$I = \frac{50 - j25}{(5 + j5) + (3 - j4)} = \frac{50 - j25}{8 + j1}$$

$$V_{th} = 50 - I(5 + j5) = \frac{25 - j75}{8 + j1} = 1.923 - j9.615 = 9.806\angle -78.69°\ \text{V}$$

$$Z_{th} = \frac{(5 + j5)(3 - j4)}{8 + j1} = \frac{35 - j5}{8 + j1} = 4.231 - j1.154\ \Omega$$

$$Z_L = Z_{th}^{*} = 4.231 + j1.154\ \Omega, \qquad P_{max} = \frac{9.806^2}{4 \times 4.231} = \frac{96.15}{16.92} = 5.68\ W$$

**ANSWER**

**$Z_L = 4.23 + j1.154\ \Omega$, $P_{max} = 5.68$ W** (= book to rounding: it squares 9.8 and prints 5.676 W)

---

## A43 · Problem 5.43

Maximum power · equivalent from measurements **new idea**

**QUESTION**

A network has two output terminals. The open circuit voltage at these terminals is 260 V. The current flowing through the terminals is 20 A when the terminals are short circuited. Also, the current is 13 A when a coil of 11 ohm reactance and negligible resistance is connected across the terminals. Find the impedance components of the equivalent circuit feeding the terminals. What value of load impedance will give maximum power transfer and what is the value of this power?

**TRY IT FIRST**

Let $Z_{th} = R + jX$. The two current readings give two equations for the magnitudes $|R + jX|$ and $|R + j(X + 11)|$.

**HINT: WHAT'S NEW**

- **$Z_{th}$ from terminal measurements:** $|Z_{th}| = V_{oc}/I_{sc}$ gives $R^2 + X^2$; a known reactance added in series gives a second equation. Subtracting them isolates X.

$$|Z_{th}| = \frac{260}{20} = 13\ \Omega \quad \Rightarrow \quad R^2 + X^2 = 169 \quad \ldots(i)$$

$$\frac{260}{|R + j(X + 11)|} = 13 \quad \Rightarrow \quad R^2 + (X + 11)^2 = 400 \quad \ldots(ii)$$

(ii) − (i): $22X + 121 = 231 \quad \Rightarrow \quad X = 5\ \Omega, \quad R = \sqrt{(169 - 25)} = 12\ \Omega$

$$Z_L = Z_{th}^{*} = 12 - j5\ \Omega, \qquad P_{max} = \frac{260^2}{4 \times 12} = 1408.3\ W$$

**ANSWER**

**$Z_{th} = 12 + j5\ \Omega$; $Z_L = 12 - j5\ \Omega$, $P_{max} = 1408.3$ W** (= book)

**CHECK**

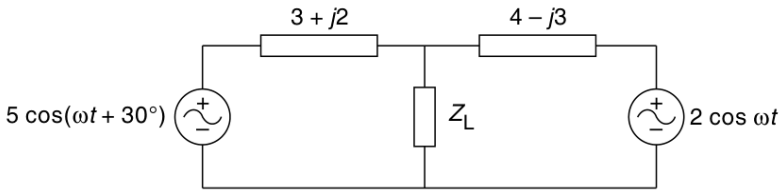
$260/|12 + j5| = 260/13 = 20$ A ✓; $260/|12 + j16| = 260/20 = 13$ A ✓.

## A44 · Problem 5.44

Maximum power · AC, two sources **practice**

**QUESTION**

What should be the value of $Z_L$ for maximum power to be delivered in the circuit shown in the figure?




Impedances are given in ohms; both sources have the same ω.

$$Z_{th} = (3 + j2) \parallel (4 - j3) = \frac{(3 + j2)(4 - j3)}{7 - j1} = \frac{18 - j1}{7 - j1} = \frac{127 + j11}{50} = 2.54 + j0.22\ \Omega$$

**ANSWER**

**$Z_L = Z_{th}^* = 2.54 - j0.22 = 2.55\angle -4.95°\ \Omega$** (= book)

**SIDE NOTE**

The source values do not affect $Z_L$; they only set $V_{th}$ and hence $P_{max}$. Thevenin's equivalent works here because both sources share one frequency ω.